Sentiment Analysis with CNN, LSTM, Transformers

In [28]:

```
!pip install matplotlib
!pip install seaborn
!pip install nltk
!pip install torch
!pip install scikit-learn
!pip install WordCloud
```

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0:00
Installing collected packages: WordCloud
Successfully installed WordCloud-1.9.4
```

In [29]:

```python
import pandas as pd
import re
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer

from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, accuracy_score, confusion
from sklearn.preprocessing import LabelEncoder
```

```
import torch
from tqdm import tqdm
from torch.utils.data import DataLoader, Dataset
from sklearn.feature_extraction.text import CountVectorizer
from collections import Counter
from wordcloud import WordCloud

# test by huggingface 开源 bert
from transformers import BertTokenizer, BertForSequenceClassification
from transformers import RobertaForSequenceClassification, RobertaTokenizer
from transformers import DistilBertForSequenceClassification, DistilBertToke
from transformers import AutoTokenizer, AutoModelForSequenceClassification

from torch.optim import AdamW
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
import pandas as pd
from collections import Counter
import re

import warnings
warnings.filterwarnings("ignore")
```

Load Data

In [31]:

```
df = pd.read_csv('data/tripadvisor_hotel_reviews.csv',
                 encoding="utf-8",
                 encoding_errors="replace")
df.head()
```

Out[31]:

	Review	Rating
0	nice hotel expensive parking got good deal sta...	4
1	ok nothing special charge diamond member hilto...	2
2	nice rooms not 4* experience hotel monaco seat...	3
3	unique, great stay, wonderful time hotel monac...	5
4	great stay great stay, went seahawk game aweso...	5

In [32]:

```
df.describe().round(2)
```

Out [32]:

	Rating
count	20491.00
mean	3.95
std	1.23
min	1.00
25%	3.00
50%	4.00
75%	5.00
max	5.00

Label sentiment by rating

In [33]:

```
# label hotel rating to label
def label_sentiment(rating):
    if rating < 3:
        return 'negative'
    elif rating >3:
        return 'positive'

df['sentiment'] = df['Rating'].apply(label_sentiment)
df.head()
```

Out [33]:

	Review	Rating	sentiment
0	nice hotel expensive parking got good deal sta...	4	positive
1	ok nothing special charge diamond member hilto...	2	negative
2	nice rooms not 4* experience hotel monaco seat...	3	None
3	unique, great stay, wonderful time hotel monac...	5	positive
4	great stay great stay, went seahawk game aweso...	5	positive

In [36]:

```
# drop neutral rating
df=df.dropna(subset=['sentiment'])
```

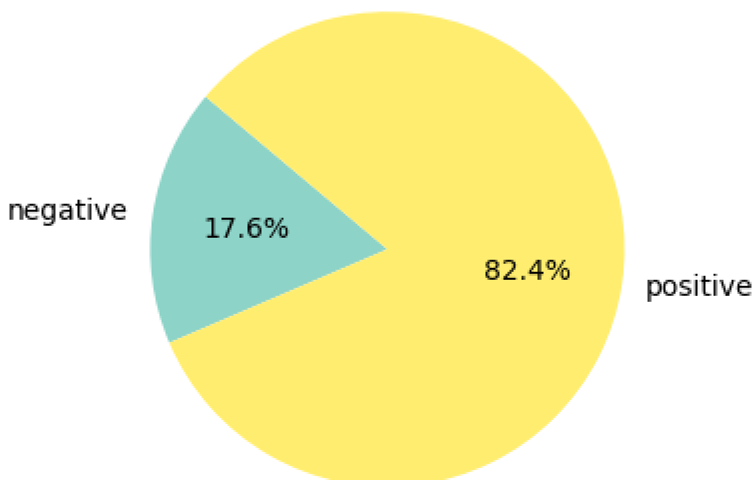
Data Visualization

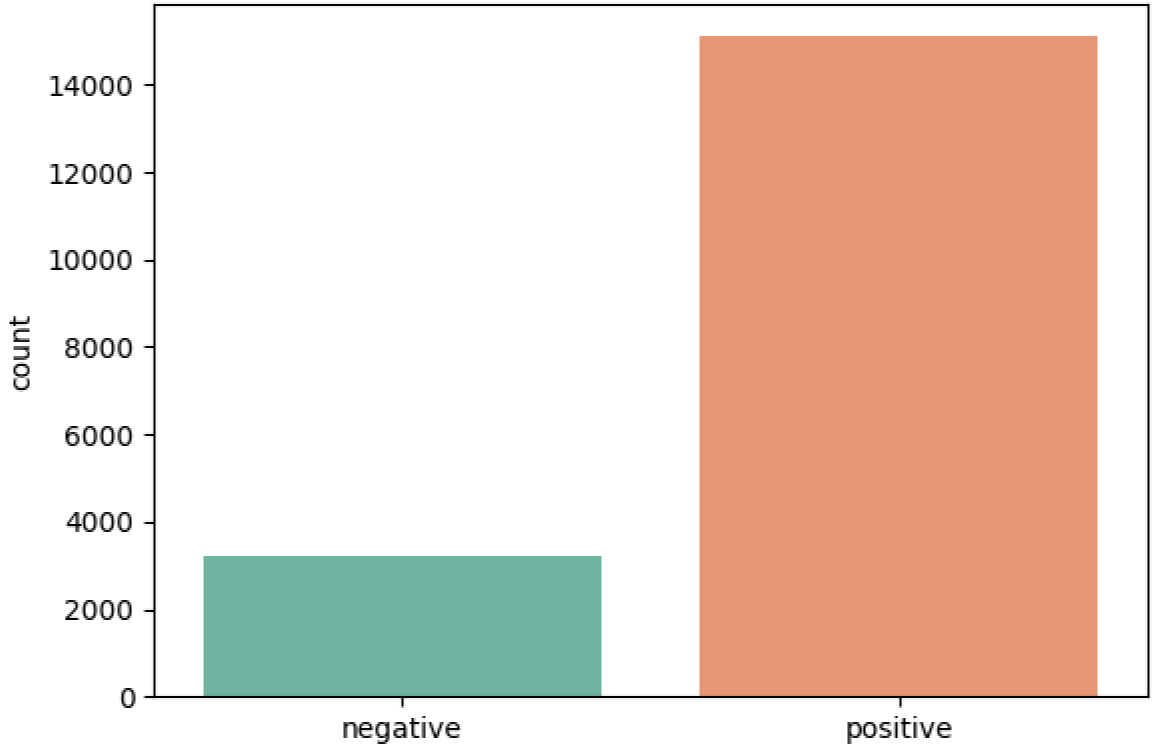
In [37]:

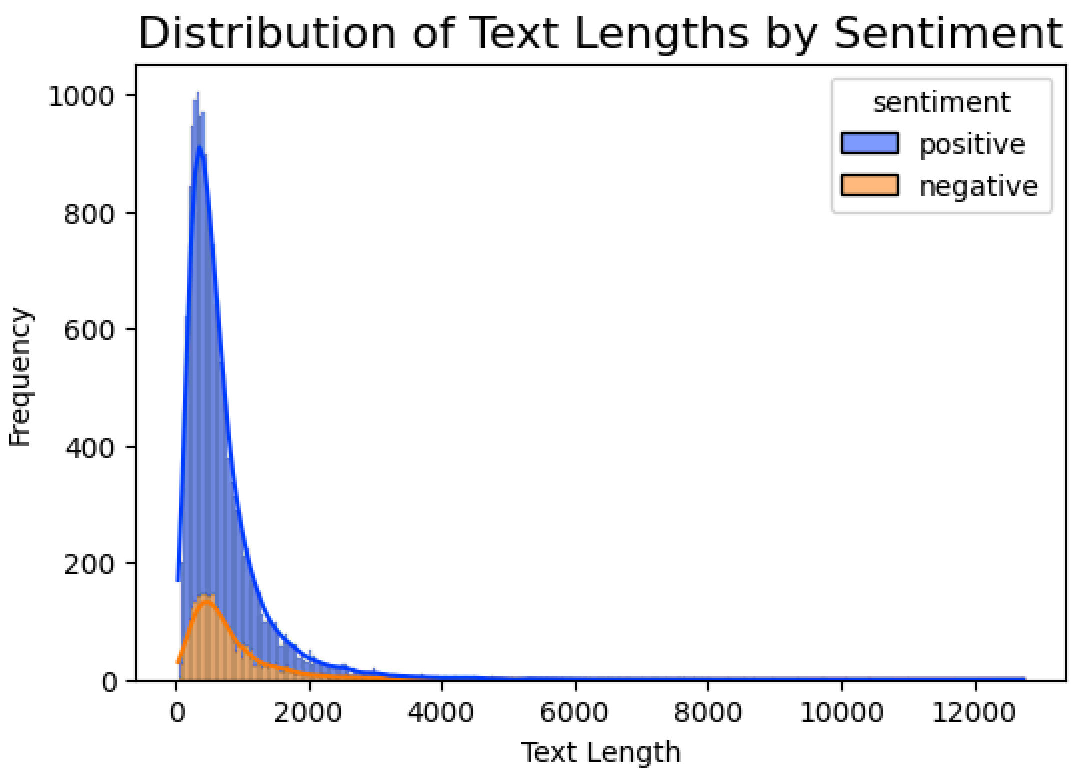
```
def draw_histogram(cat, count, title="Distribution of Ratings Histogram"):
    plt.figure(figsize=(6, 4))
    barplot = sns.barplot(
        x=cat,
        y=count,
        dodge=False,
        palette="Set2"
```

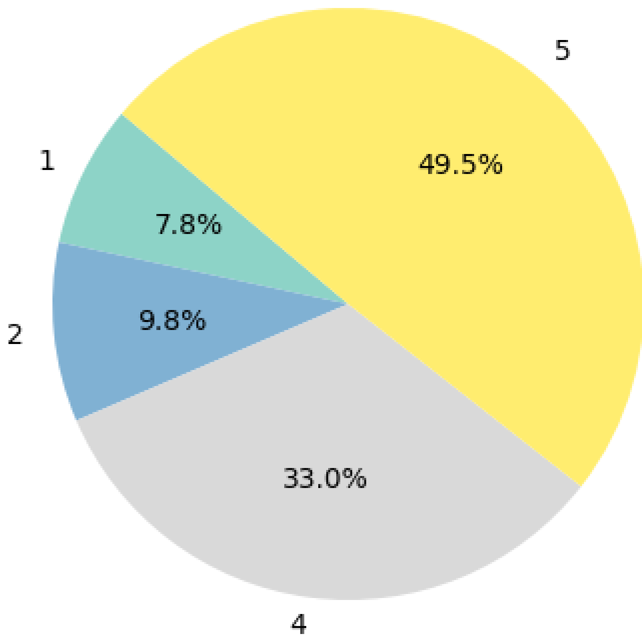
```
    )
    # plt.legend(title=title, loc='upper right')
    plt.tight_layout()
    plt.show()


def draw_pie(cat, count, title="Distribution of Ratings Bar"):
    plt.figure(figsize=(4, 4))
    colors = plt.cm.Set3(np.linspace(0, 1, len(cat)))  # automatically fill
    plt.pie(
        count,
        labels=cat,
        autopct="%.1f%%",
        colors=colors,
        startangle=140
    )
    plt.tight_layout()
    plt.show()

def word_length_dist(df, x='word_length'):
    # Calculate the length of each review
    plt.figure(figsize=(6, 4))
    sns.histplot(df, x='word_length', hue='sentiment', multiple='stack', pal
    plt.title('Distribution of Text Lengths by Sentiment', fontsize=16)
    plt.xlabel('Text Length')
    plt.ylabel('Frequency')
    plt.show()


def show_wordcloud(reviews, title="Word Cloud", filter_words = None):
    wc = WordCloud(
        max_words=500,
        background_color="white",
        stopwords=filter_words).generate(" ".join(reviews))

    plt.figure(figsize=(6, 4))
    plt.imshow(wc)
    plt.title(title, fontsize=20)
    plt.axis('off')
    plt.show()

def word_count(df):
    vectorizer = CountVectorizer(stop_words='english', max_features=10)
    word_counts = vectorizer.fit_transform(df['Review']).toarray().sum(axis=
    words = vectorizer.get_feature_names_out()
    word_freq = dict(zip(words, word_counts))
    sorted_word_freq = dict(sorted(word_freq.items(), key=lambda item: item[

    # Plot the most common words
    plt.figure(figsize=(6, 4))
    sns.barplot(x=list(sorted_word_freq.keys()), y=list(sorted_word_freq.val
    plt.title('Most Common Words')
    plt.xlabel('Words')
    plt.ylabel('Frequency')
    plt.xticks(rotation=45)
    plt.show()
```

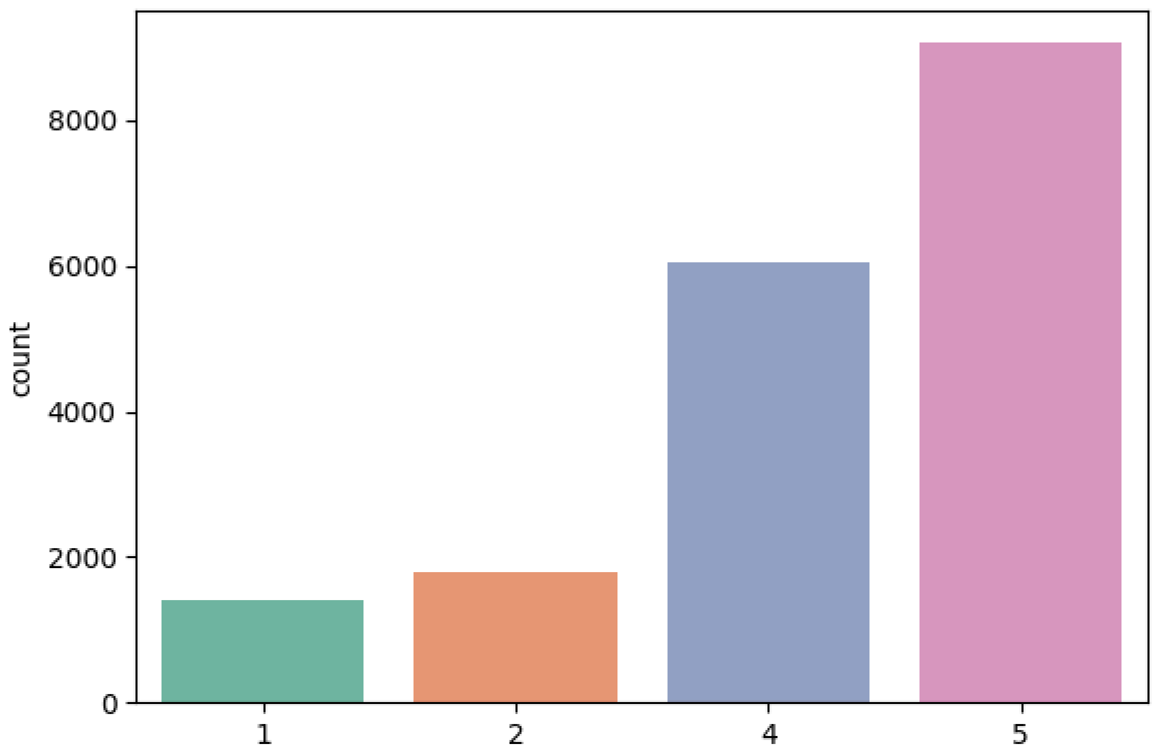
In [38]:
```
# visualization
df['word_length'] = df['Review'].apply(len)

sentiment_rate = df['sentiment'].value_counts(normalize=True).sort_index()

colors = sns.color_palette("viridis", len(sentiment_rate))
labels = [f"{label}" for label in sentiment_rate.index]

draw_pie(labels, sentiment_rate)
draw_histogram(labels, df['sentiment'].value_counts().sort_index())
word_length_dist(df)


# rating
ratings_count = df['Rating'].value_counts(normalize=True).mul(100).sort_inde
rates = [f"{rate}" for rate in ratings_count.index]

draw_pie(rates, ratings_count)
draw_histogram(rates, df['Rating'].value_counts().sort_index())
```



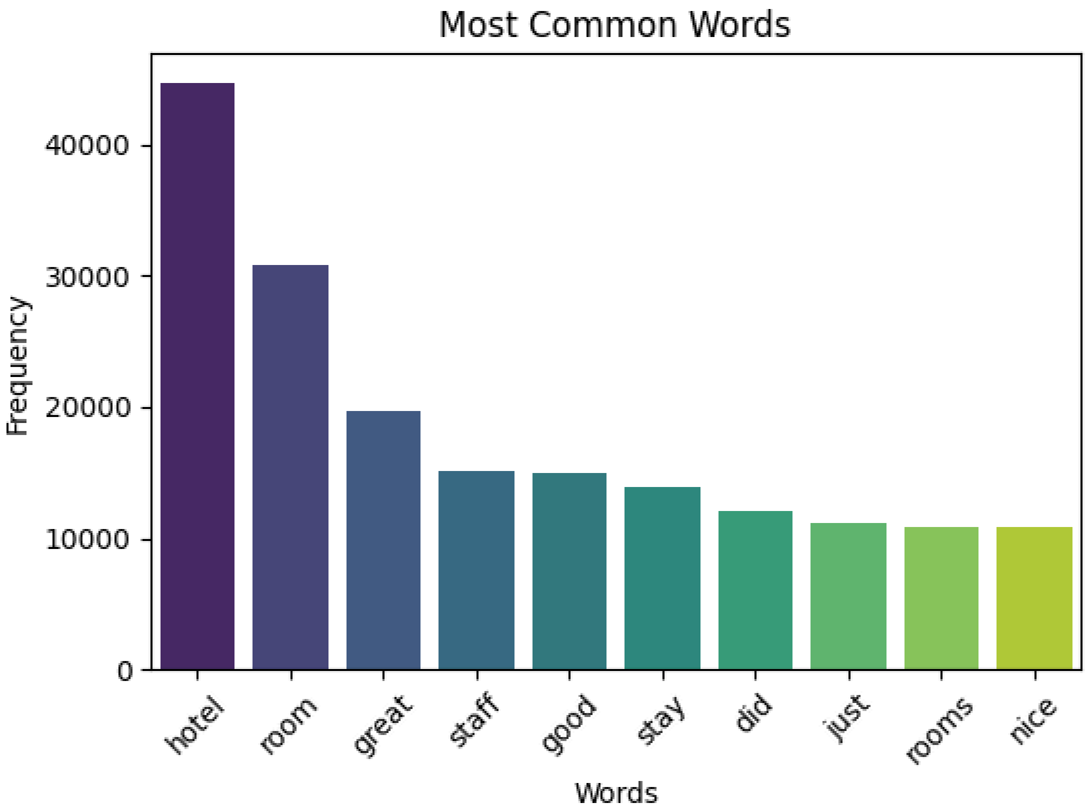

14000
12000
10000
8000
6000
4000
2000
0
count
negative
positive

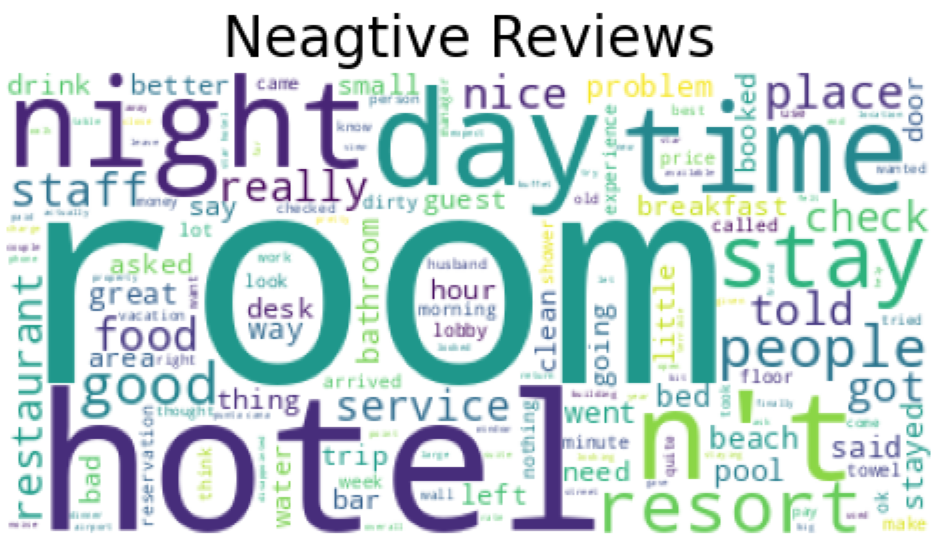

Distribution of Text Lengths by Sentiment
sentiment
positive
negative
1000
800
600
400
200
0
Frequency
0
2000
4000
6000
8000
10000
12000
Text Length





In [39]:
```
word_count(df)

show_wordcloud(df[df["sentiment"] == "negative"]["Review"], title="Neagtive 
show_wordcloud(df[df["sentiment"] == "positive"]["Review"], title="Postive R
```


Most Common Words
Frequency
0
10000
20000
30000
40000
hotel
room
great
staff
good
stay
did
just
rooms
nice
Words


Neagtive Reviews
room
hotel
night
day
time
stay
resort
n't
staff
people
place
nice
really
good
food
service
restaurant
told
check
got
breakfast
problem
bathroom
clean
going
little
beach
pool
said
stayed
booked
door
small
better
drink
great
asked
desk
way
hour
lobby
morning
area
thing
trip
water
bar
left
went
need
bed
floor
guest
dirty
called
say
lot
bad
think
reservation
arrived
minute
nothing
towel
make

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 18307 entries, 0 to 20490
Data columns (total 4 columns):
 #   Column       Non-Null Count  Dtype 
---  ------       --------------  ----- 
 0   Review       18307 non-null  object
 1   Rating       18307 non-null  int64 
 2   sentiment    18307 non-null  object
 3   word_length  18307 non-null  int64 
dtypes: int64(2), object(2)
memory usage: 715.1+ KB
```

Text Preprocessing Pipeline

In [13]:
```
# Download required NLTK data
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\HW\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\HW\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\HW\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\HW\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
```

Out[13]: True

In [ ]:
```
def preprocess_data(df_data):
    """
    Preprocesses a DataFrame by cleaning and preparing text data in the 'Rev

    Steps:
    1. Remove duplicates and rows with missing values.
    2. Clean 'Review' column:
       - Strip whitespace.
       - Remove non-alphabetic characters (except spaces).
       - Replace newlines with spaces.
       - Convert to lowercase.
       - Remove URLs.
       - Normalize spaces.
    3. Remove rows with empty 'Review' after cleaning.
    4. Reset index.

    Args:
        df_data (pd.DataFrame): Input DataFrame with 'Review' and 'Rating' c
    Returns:
        pd.DataFrame: Cleaned DataFrame.
    """
    # df = df_data.dropna().copy()
    # print(df_data.shape)
    # print(df.shape)

    df = df_data.copy()
    df['Review'] = (df_data['Review']
        .str.strip()
        .str.replace(r'\n', ' ', regex=True)
        .str.lower()
        .apply(lambda x: re.sub(r'http\S+|www.\S+', '', x))
        .apply(lambda x: re.sub(r'[^a-z\s]', '', x))
        .apply(lambda x: ' '.join(x.split()))
    )
    df = df[df['Review'].str.len() > 0].reset_index(drop=True)
    df = df[df['word_length'] <= 2000]
    return df

# print(df['Review'].isna().sum())
# print(df['sentiment'].isna().sum())
# df[df[['Review','airline']].duplicated()]
# new_df = preprocess_data(df)
# display(df.shape)
# display(new_df.shape)

def tokenize_text(df):
    """
    Tokenizes the text in the 'Review' column of the DataFrame.

    Args:
        df (pd.DataFrame): DataFrame containing a 'Review' column.

    Returns:
        pd.DataFrame: DataFrame with a new 'tokens' column containing tokeni
    """
```

```
    df['tokens'] = df['Review'].apply(lambda review: word_tokenize(review))
    return df

def remove_stopwords(df):
    """
    Removes stopwords from the tokenized text in the 'tokens' column.

    Args:
        df (pd.DataFrame): DataFrame with a 'tokens' column.

    Returns:
        pd.DataFrame: DataFrame with a new 'tokens_no_stop' column containir
    """
    stop_words = set(stopwords.words('english'))
    df['tokens_no_stop'] = df['tokens'].apply(
        lambda tokens: [word for word in tokens if word.lower() not in stop_
    )
    return df

# pos_tags = pos_tag(tokens)  # get speech
# lemmatized = [lemmatizer.lemmatize(word, get_wordnet_pos(pos)) for word, p
def lemmatize_text(df):
    """
    Lemmatizes the tokens in the 'tokens_no_stop' column.

    Args:
        df (pd.DataFrame): DataFrame with a 'tokens_no_stop' column.

    Returns:
        pd.DataFrame: DataFrame with a new 'lemmatized' column containing le
    """
    lemmatizer = WordNetLemmatizer()
    df['lemmatized'] = df['tokens_no_stop'].apply(
        lambda tokens: [lemmatizer.lemmatize(word) for word in tokens]
    )
    return df

def join_tokens(df):
    """
    Joins the lemmatized tokens back into a single string.

    Args:
        df (pd.DataFrame): DataFrame with a 'lemmatized' column.
    Returns:
        pd.DataFrame: DataFrame with a new 'processed_text' column containir
    """
    df['processed_text'] = df['lemmatized'].apply(
        lambda tokens: ' '.join(tokens)
    )
    return df
```

In [15]:

```
df = (
    preprocess_data(df)
    .pipe(tokenize_text)
    .pipe(remove_stopwords)
    .pipe(lemmatize_text)
```

```
    .pipe(join_tokens)
)
```

In [16]:
```
# check data shape, postive/negative rate after preprocess

df = df[df['sentiment'].isin(["positive", "negative"])]
display(df.head(3))
print(df.shape)

print(df['processed_text'].isna().sum())
print(df['sentiment'].isna().sum())


sentiment_rate = df['sentiment'].value_counts(normalize=True).sort_index()
sentiment_rate
```

	Review	Rating	sentiment	word_length	tokens	tokens_no_stop	lemmatized
0	nice hotel expensive parking got good deal sta...	4	positive	593	[nice, hotel, expensive, parking, got, good, d...	[nice, hotel, expensive, parking, got, good, d...	[nice, hotel, expensive, parking, got, good, d...
1	ok nothing special charge diamond member hilto...	2	negative	1689	[ok, nothing, special, charge, diamond, member...	[ok, nothing, special, charge, diamond, member...	[ok, nothing, special, charge, diamond, member...
2	unique great stay wonderful time hotel monaco ...	5	positive	600	[unique, great, stay, wonderful, time, hotel, ...	[unique, great, stay, wonderful, time, hotel, ...	[unique, great, stay, wonderful, time, hotel, ...

```
(17515, 8)
0
0
```

Out[16]:
```
sentiment
negative    0.172538
positive    0.827462
Name: proportion, dtype: float64
```

Sentiment Analysis Model

In [17]:
```
sentiment_rate = df['sentiment'].value_counts(normalize=True).sort_index()
labels = [f"{label}" for label in sentiment_rate.index]
labels
```

Out[17]: ['negative', 'positive']

In [ ]:
```
def plot_confusion_matrix(y_true, y_pred, labels=None, cmap="Blues"):
    cm = confusion_matrix(y_true, y_pred)
    cm = cm.astype("float") / cm.sum(axis=1, keepdims=True)
    plt.figure(figsize=(6, 5))
    sns.heatmap(cm, annot=True, fmt=".2f", cmap=cmap, xticklabels=labels, yt
    plt.xlabel("Predicted")
    plt.ylabel("Actual")
    plt.title("Normalized Confusion Matrix")
    plt.show()


def plot_auc(pred_nb, y_test):
    from sklearn.metrics import auc, f1_score, precision_recall_curve, recal

    fpr, tpr, _ = roc_curve(y_test, pred_nb)
    precision, recall, _ = precision_recall_curve(y_test, pred_nb)

    roc_auc = auc(fpr, tpr)
    pr_auc = auc(recall, precision)

    plt.plot(fpr, tpr, label='ROC Curve (AUC = {:.5f})'.format(roc_auc))
    plt.plot(recall, precision, label='PR Curve (AUC = {:.5f})'.format(pr_au


    plt.plot([0, 1], [0, 1], linestyle='--', label='Random', color='gray')

    # setting axis ,limits and label
    plt.xlim([0.0, 1.0])
    plt.ylim([0.0, 1.05])
    plt.xlabel('False Positive Rate / Recall')
    plt.ylabel('True Positive Rate / Precision')
    plt.title('PR-ROC ')
    plt.legend(loc='lower right')
    # plt.savefig('PR-ROC Curve.png')

    plt.show()

def model_eval(model_name, y_pred, y_test):
    print(f"{model_name} Accuracy: {accuracy_score(y_test, y_pred) * 100:.2f
    report = classification_report(y_test, y_pred, zero_division=0)
    print(report)
    plot_confusion_matrix(y_test, y_pred, labels)
    return report

def draw_bar(acc, pre, recall):

    metrics = ['Accuracy', 'Precision', 'Recall']
    values = [acc, pre, recall]

    plt.figure(figsize=(8, 6))
    bars = plt.bar(metrics, values, color=['blue', 'green', 'red'])

    for bar in bars:
```

```
        height = bar.get_height()
        plt.text(bar.get_x() + bar.get_width()/2., height,
                f'{height:.2f}',
                ha='center', va='bottom')

    plt.title('Classification Metrics', fontsize=14)
    plt.ylabel('Score', fontsize=12)
    plt.ylim(0, 1.1)

    plt.grid(axis='y', linestyle='--', alpha=0.7)
    plt.tight_layout()
    plt.show()
```

Transformer

In [67]:
```
import torch
print(torch.cuda.is_available())
print(torch.version.cuda)
print(torch.cuda.get_device_name(0))
```

```
True
12.1
NVIDIA GeForce RTX 4060 Laptop GPU
```

In [21]:
```
class SentimentDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_len=512):
        self.texts = texts
        self.labels = labels
        self.tokenizer = tokenizer
        self.max_len = max_len

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, item):
        texts = str(self.texts[item])
        labels = self.labels[item]

        # text to vector
        encoding = self.tokenizer.encode_plus(
            texts,
            add_special_tokens=True,
            max_length=self.max_len,
            return_token_type_ids=False,
            padding='max_length',
            truncation=True,
            return_attention_mask=True,
            return_tensors='pt'
        )

        input_ids = encoding['input_ids'].flatten()
        attention_mask = encoding['attention_mask'].flatten()

        return {
```

```
            'input_ids': input_ids,
            'attention_mask': attention_mask,
            'labels': torch.tensor(labels, dtype=torch.long)
        }
```

In [22]:
```
from transformers import DataCollatorWithPadding
def create_dataloaders(train_df, test_df, tokenizer, label = 'sentiment', ba
    data_collator = DataCollatorWithPadding(tokenizer)
    # train_loader = DataLoader(dataset, batch_size=8, collate_fn=data_colla

    train_dataset = SentimentDataset(train_df["processed_text"].tolist(), tr
    test_dataset = SentimentDataset(test_df["processed_text"].tolist(), test

    train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=
    test_loader = DataLoader(test_dataset, batch_size=batch_size, shuffle=Fa
    return train_loader, test_loader
```

In [ ]:
```
def train_epoch(model, optimizer, device, train_loader, epoch):
    model.train()
    running_loss = 0.0
    correct = 0
    total = 0

    optimizer.zero_grad()
    progress_bar = tqdm(train_loader, desc=f"训练中 {epoch}")
    for i, batch in enumerate(progress_bar):
        input_ids, attention_mask, labels = (
            batch[k].to(device) for k in ["input_ids", "attention_mask", "la
        )
        # print(f"label = {labels}")
        outputs = model(input_ids, attention_mask=attention_mask, labels=lab
        loss = outputs.loss
        loss.backward()

        # Gradient clipping to avoid gradient explosion
        # nn.utils.clip_grad_norm_(self.model.parameters(), max_norm=5.0)

        optimizer.step()
        optimizer.zero_grad()

        # total_loss += loss.item()


        running_loss += loss.item()
        # _, predicted = outputs.max(1)
        predicted = torch.argmax(outputs.logits, dim=1)
        total += labels.size(0)
        correct += predicted.eq(labels).sum().item()

        # update progress bar
        progress_bar.set_postfix({
            'loss': running_loss / (progress_bar.n + 1),
            'acc': 100. * correct / total
        })
```

```
    epoch_loss = running_loss / len(train_loader)
    epoch_acc = 100. * correct / total
    return epoch_loss, epoch_acc

from sklearn.metrics import accuracy_score
def evaluate2(model, device, test_loader):
    model.eval()
    total_loss = 0
    all_preds = []
    all_labels = []

    with torch.no_grad():
        for batch in tqdm(test_loader, desc="Evaluating"):
            input_ids, attention_mask, labels = (
                batch[k].to(device) for k in ["input_ids", "attention_mask",
            )

            outputs = model(input_ids, attention_mask=attention_mask, labels
            loss = outputs.loss
            logits = outputs.logits

            total_loss += loss.item()
            preds = torch.argmax(logits, dim=1)
            all_preds.extend(preds.cpu().numpy())
            all_labels.extend(labels.cpu().numpy())

    avg_loss = total_loss / len(test_loader)
    acc = accuracy_score(all_labels, all_preds)

    return all_preds, all_labels, avg_loss, acc*100.


def train(model,
          train_loader,
          test_loader,
          num_epochs=3,
          # learning_rate=2e-5,
          learning_rate=0.001,
          save_path="model.pth",
          target_names = ["negative", "neutral", "positive"],
          fine_tuning = False):

     # move to the specified device
    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    model.to(device)

    history = {}
    train_losses = []
    train_accuracies = []
    val_losses = []
    val_accuracies = []

    if fine_tuning:
        for layer in model.distilbert.transformer.layer[:4]:
            for param in layer.parameters():
                param.requires_grad = False
```

```
        optimizer = torch.optim.AdamW(model.classifier.parameters(), lr=lear
    else:
        optimizer = AdamW(model.parameters(), lr=learning_rate)

    scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(optimizer, 'min',

    for epoch in range(1, num_epochs + 1):
        # train
        train_loss, train_acc = train_epoch(model, optimizer, device, train_
        train_losses.append(train_loss)
        train_accuracies.append(train_acc)

        test_preds, test_true,  val_loss, val_acc = evaluate2(model, device,
        val_losses.append(val_loss)
        val_accuracies.append(val_acc)

        # adjust lr
        scheduler.step(train_loss)

        history = {
            'train_losses': train_losses,
            'train_accuracies': train_accuracies,
            'val_losses': val_losses,
            'val_accuracies': val_accuracies
        }
        print(
            "Classification Report:\n",
            classification_report(
                test_true, test_preds, target_names=target_names
            ),
        )

    # Save model
    print(f"\nSaving model...{save_path}")
    torch.save(
        {
            "model_state_dict": model.state_dict(),
            "optimizer_state_dict": optimizer.state_dict()
        },
        save_path,
    )
    return history
```

In [30]:
```
# distilbert-base-uncased
tokenizer_hotel = AutoTokenizer.from_pretrained('distilbert-base-uncased')
model_hotel = AutoModelForSequenceClassification.from_pretrained('distilbert
```

```
Some weights of DistilBertForSequenceClassification were not initialized fro
m the model checkpoint at distilbert-base-uncased and are newly initialized:
['classifier.bias', 'classifier.weight', 'pre_classifier.bias', 'pre_classif
ier.weight']
You should probably TRAIN this model on a down-stream task to be able to use
it for predictions and inference.
```

In [ ]:
```
# encode label
df['sentiment_encode'] = LabelEncoder().fit_transform(df['sentiment'].dropna
```

```python
print(df['sentiment_encode'].value_counts())

train_df_hotel, test_df_hotel = train_test_split(
    df, test_size=0.1, random_state=42, stratify=df['sentiment_encode']
)

train_loader_hotel, test_loader_hotel = create_dataloaders(train_df_hotel, t
                                                           tokenizer_hotel,
                                                           batch_size = 16)
```

```
sentiment_encode
1    14493
0     3022
Name: count, dtype: int64
```

In [ ]:

```python
history_hotel = train(model_hotel,
      train_loader_hotel,
      test_loader_hotel,
      save_path="distil_bert_hotel.pth",
      fine_tuning = True,
      # target_names = ['negative', 'neutral', 'positive'],
      target_names = ['negative', 'positive'],
      num_epochs=50
     )
```

```
训练中 1: 100%|██████████| 2060/2060 [01:16<00:00, 27.02it/s, loss=0.248, acc
=89.8]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.09it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.92      0.57      0.71       321
    positive       0.92      0.99      0.95      1510

    accuracy                           0.92      1831
   macro avg       0.92      0.78      0.83      1831
weighted avg       0.92      0.92      0.91      1831

训练中 2: 100%|██████████| 2060/2060 [01:16<00:00, 26.87it/s, loss=0.244, acc
=90.5]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.88it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.61      0.72       321
    positive       0.92      0.98      0.95      1510

    accuracy                           0.92      1831
   macro avg       0.91      0.80      0.84      1831
weighted avg       0.92      0.92      0.91      1831

训练中 3: 100%|██████████| 2060/2060 [01:15<00:00, 27.29it/s, loss=0.243, acc
=90.1]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.68it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.95      0.51      0.66       321
    positive       0.90      0.99      0.95      1510

    accuracy                           0.91      1831
   macro avg       0.93      0.75      0.81      1831
weighted avg       0.91      0.91      0.90      1831
```

```
训练中 4: 100%|██████████| 2060/2060 [01:16<00:00, 26.86it/s, loss=0.243, acc
=90]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.17it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.88      0.67      0.76       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.92      0.93      0.92      1831
```

```
训练中 5: 100%|██████████| 2060/2060 [01:16<00:00, 26.88it/s, loss=0.237, acc
=90.4]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.93it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.88      0.69      0.78       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.84      0.87      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 6: 100%|██████████| 2060/2060 [01:16<00:00, 26.93it/s, loss=0.236, acc
=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.07it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.82      0.76      0.79       321
    positive       0.95      0.96      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.88      0.86      0.87      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 7: 100%|██████████| 2060/2060 [01:17<00:00, 26.67it/s, loss=0.238, acc
=90.1]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.70it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.66      0.76       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.92      0.82      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 8: 100%|██████████| 2060/2060 [01:16<00:00, 26.92it/s, loss=0.237, acc
=90.3]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.27it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.91      0.60      0.72       321
    positive       0.92      0.99      0.95      1510

    accuracy                           0.92      1831
   macro avg       0.91      0.79      0.84      1831
weighted avg       0.92      0.92      0.91      1831
```

```
训练中 9: 100%|██████████| 2060/2060 [01:16<00:00, 26.97it/s, loss=0.235, acc
=90.5]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.22it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.87      0.70      0.78       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.90      0.84      0.87      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 10: 100%|██████████| 2060/2060 [01:16<00:00, 26.97it/s, loss=0.238, ac
c=90.3]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.10it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.66      0.76       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.82      0.86      1831
weighted avg       0.92      0.93      0.92      1831
```

```
训练中 11: 100%|██████████| 2060/2060 [01:16<00:00, 26.96it/s, loss=0.24, acc
=90.1]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.04it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.64      0.74       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.92      1831
   macro avg       0.91      0.81      0.85      1831
weighted avg       0.92      0.92      0.92      1831
```

```
训练中 12: 100%|██████████| 2060/2060 [01:16<00:00, 26.95it/s, loss=0.239, ac
c=90.4]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.37it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.64      0.74       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.92      1831
   macro avg       0.91      0.81      0.85      1831
weighted avg       0.92      0.92      0.92      1831
```

```
训练中 13: 100%|██████████| 2060/2060 [01:16<00:00, 26.97it/s, loss=0.235, ac
c=90.4]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.08it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.88      0.69      0.77       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 14: 100%|██████████| 2060/2060 [01:16<00:00, 26.86it/s, loss=0.235, ac
c=90.6]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.03it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.62      0.74       321
    positive       0.92      0.99      0.95      1510

    accuracy                           0.92      1831
   macro avg       0.91      0.80      0.84      1831
weighted avg       0.92      0.92      0.92      1831
```

```
训练中 15: 100%|██████████| 2060/2060 [01:16<00:00, 26.82it/s, loss=0.232, ac
c=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.78it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.92      0.60      0.73       321
    positive       0.92      0.99      0.95      1510

    accuracy                           0.92      1831
   macro avg       0.92      0.80      0.84      1831
weighted avg       0.92      0.92      0.91      1831

训练中 16: 100%|██████████| 2060/2060 [01:16<00:00, 26.84it/s, loss=0.237, ac
c=90.5]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.85it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.68      0.77       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831

训练中 17: 100%|██████████| 2060/2060 [01:16<00:00, 26.82it/s, loss=0.233, ac
c=90.6]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.84it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.88      0.69      0.77       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.87      1831
weighted avg       0.93      0.93      0.93      1831

训练中 18: 100%|██████████| 2060/2060 [01:16<00:00, 27.00it/s, loss=0.231, ac
c=90.6]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.24it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.87      0.70      0.78       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.90      0.84      0.87      1831
weighted avg       0.93      0.93      0.93      1831

训练中 19: 100%|██████████| 2060/2060 [01:16<00:00, 26.90it/s, loss=0.233, ac
c=90.5]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.20it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.87      0.70      0.78       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.84      0.87      1831
weighted avg       0.93      0.93      0.93      1831

训练中 20: 100%|██████████| 2060/2060 [01:16<00:00, 26.94it/s, loss=0.235, ac
c=90.5]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.94it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.91      0.62      0.73       321
    positive       0.92      0.99      0.95      1510

    accuracy                           0.92      1831
   macro avg       0.92      0.80      0.84      1831
weighted avg       0.92      0.92      0.92      1831

训练中 21: 100%|██████████| 2060/2060 [01:16<00:00, 26.83it/s, loss=0.235, ac
c=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.96it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.68      0.77       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831

训练中 22: 100%|██████████| 2060/2060 [01:16<00:00, 26.83it/s, loss=0.234, ac
c=90.8]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.80it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.63      0.74       321
    positive       0.93      0.99      0.95      1510

    accuracy                           0.92      1831
   macro avg       0.91      0.81      0.85      1831
weighted avg       0.92      0.92      0.92      1831

训练中 23: 100%|██████████| 2060/2060 [01:16<00:00, 26.87it/s, loss=0.235, ac
c=90.4]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.82it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.71      0.78       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.90      0.84      0.87      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 24: 100%|██████████| 2060/2060 [01:16<00:00, 26.95it/s, loss=0.23, acc=90.8]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.30it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.90      0.64      0.75       321
    positive       0.93      0.99      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.92      0.81      0.85      1831
weighted avg       0.92      0.93      0.92      1831
```

```
训练中 25: 100%|██████████| 2060/2060 [01:16<00:00, 26.82it/s, loss=0.232, acc=90.6]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.73it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.88      0.69      0.77       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.87      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 26: 100%|██████████| 2060/2060 [01:16<00:00, 26.79it/s, loss=0.233, acc=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.71it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.90      0.66      0.76       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.82      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 27: 100%|██████████| 2060/2060 [01:16<00:00, 27.02it/s, loss=0.231, acc=90.6]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.17it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.67      0.77       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 28: 100%|██████████| 2060/2060 [01:16<00:00, 26.88it/s, loss=0.231, ac
c=90.8]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.76it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.69      0.77       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.87      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 29: 100%|██████████| 2060/2060 [01:16<00:00, 26.88it/s, loss=0.231, ac
c=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.20it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.69      0.78       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.87      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 30: 100%|██████████| 2060/2060 [01:16<00:00, 27.01it/s, loss=0.23, acc
=90.5]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.31it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.62      0.74       321
    positive       0.92      0.99      0.95      1510

    accuracy                           0.92      1831
   macro avg       0.91      0.80      0.84      1831
weighted avg       0.92      0.92      0.92      1831
```

```
训练中 31: 100%|██████████| 2060/2060 [01:16<00:00, 27.05it/s, loss=0.232, ac
c=90.4]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.06it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.90      0.64      0.75       321
    positive       0.93      0.99      0.96      1510

    accuracy                           0.92      1831
   macro avg       0.92      0.81      0.85      1831
weighted avg       0.92      0.92      0.92      1831
```

```
训练中 32: 100%|██████████| 2060/2060 [01:16<00:00, 27.02it/s, loss=0.23, acc
=90.5]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.10it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.90      0.67      0.76       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 33: 100%|██████████| 2060/2060 [01:16<00:00, 27.06it/s, loss=0.232, ac
c=90.5]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.05it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.90      0.68      0.77       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.92      0.83      0.87      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 34: 100%|██████████| 2060/2060 [01:16<00:00, 27.05it/s, loss=0.23, acc
=90.6]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.15it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.70      0.77       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.90      0.84      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 35: 100%|██████████| 2060/2060 [01:16<00:00, 27.04it/s, loss=0.23, acc
=90.9]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.15it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.67      0.76       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831

```

```
训练中 36: 100%|██████████| 2060/2060 [01:16<00:00, 27.02it/s, loss=0.23, acc=90.9]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.87it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.63      0.74       321
    positive       0.93      0.99      0.95      1510

    accuracy                           0.92      1831
   macro avg       0.91      0.81      0.85      1831
weighted avg       0.92      0.92      0.92      1831

```

```
训练中 37: 100%|██████████| 2060/2060 [01:16<00:00, 26.87it/s, loss=0.229, acc=90.8]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.26it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.68      0.77       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.92      0.83      0.86      1831
weighted avg       0.93      0.93      0.93      1831

```

```
训练中 38: 100%|██████████| 2060/2060 [01:16<00:00, 27.08it/s, loss=0.234, acc=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.35it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.67      0.76       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831

```

```
训练中 39: 100%|██████████| 2060/2060 [01:16<00:00, 27.01it/s, loss=0.234, acc=90.6]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.28it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.67      0.77       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 40: 100%|██████████| 2060/2060 [01:16<00:00, 26.99it/s, loss=0.23, acc
=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.24it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.67      0.77       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 41: 100%|██████████| 2060/2060 [01:16<00:00, 27.01it/s, loss=0.231, ac
c=90.6]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.11it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.67      0.77       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 42: 100%|██████████| 2060/2060 [01:16<00:00, 27.08it/s, loss=0.229, ac
c=90.6]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.21it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.66      0.76       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.82      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 43: 100%|██████████| 2060/2060 [01:16<00:00, 26.93it/s, loss=0.229, ac
c=90.9]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.01it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.89      0.69      0.77       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.87      1831
weighted avg       0.93      0.93      0.93      1831
```

```
训练中 44: 100%|██████████| 2060/2060 [01:16<00:00, 27.05it/s, loss=0.228, ac
c=90.9]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.20it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.90      0.64      0.75       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.92      1831
   macro avg       0.91      0.81      0.85      1831
weighted avg       0.92      0.92      0.92      1831
```

```
训练中 45: 100%|██████████| 2060/2060 [01:16<00:00, 27.04it/s, loss=0.228, ac
c=90.9]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.25it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.90      0.64      0.75       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.81      0.85      1831
weighted avg       0.92      0.93      0.92      1831
```

```
训练中 46: 100%|██████████| 2060/2060 [01:16<00:00, 27.03it/s, loss=0.231, ac
c=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 27.99it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.90      0.67      0.77       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.92      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831
```

```
训练中 47: 100%|██████████| 2060/2060 [01:16<00:00, 27.08it/s, loss=0.231, ac
c=90.8]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.14it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.64      0.75       321
    positive       0.93      0.99      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.92      0.82      0.85      1831
weighted avg       0.92      0.93      0.92      1831

训练中 48: 100%|██████████| 2060/2060 [01:16<00:00, 27.04it/s, loss=0.23, acc
=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.24it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.68      0.77       321
    positive       0.94      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.83      0.87      1831
weighted avg       0.93      0.93      0.93      1831

训练中 49: 100%|██████████| 2060/2060 [01:16<00:00, 27.05it/s, loss=0.232, ac
c=90.7]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.24it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.66      0.76       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.91      0.82      0.86      1831
weighted avg       0.93      0.93      0.92      1831

训练中 50: 100%|██████████| 2060/2060 [01:16<00:00, 26.98it/s, loss=0.233, ac
c=90.5]
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.20it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.67      0.77       321
    positive       0.93      0.98      0.96      1510

    accuracy                           0.93      1831
   macro avg       0.92      0.83      0.86      1831
weighted avg       0.93      0.93      0.92      1831


Saving model...distil_bert_hotel.pth
```

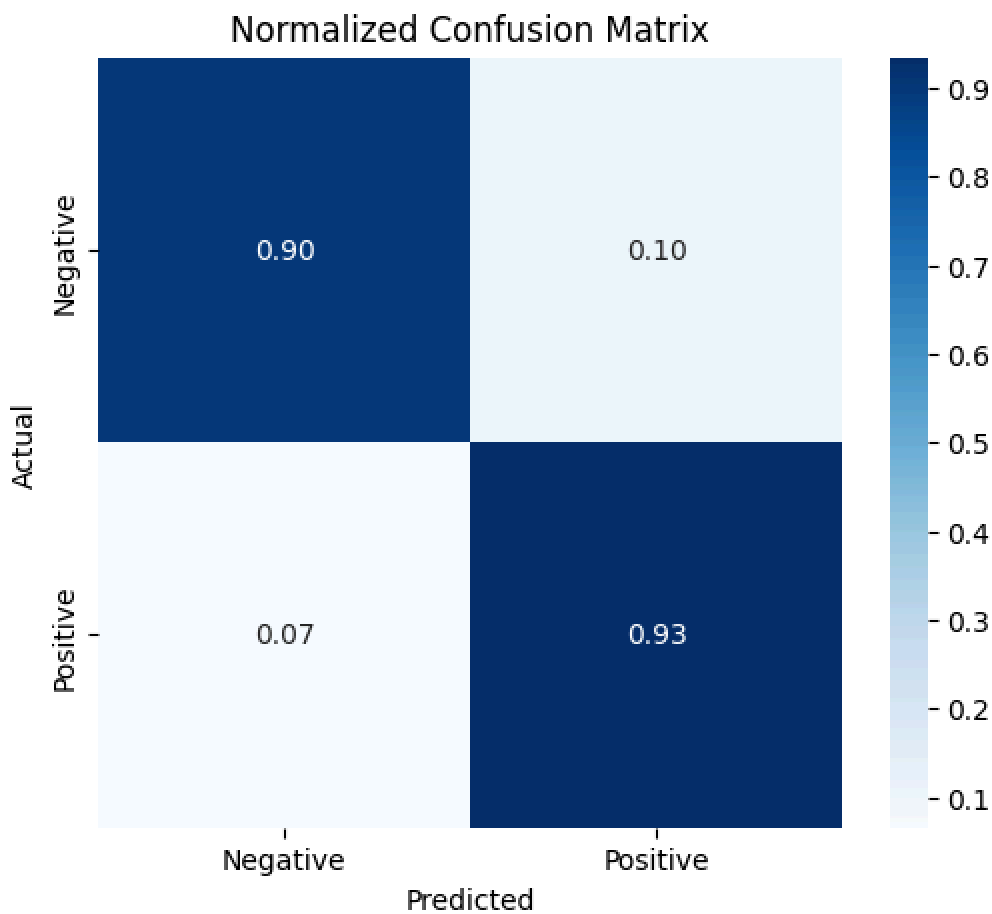
In [ ]:

```python
checkpoint = torch.load("distil_bert_hotel.pth")

model_hotel = DistilBertForSequenceClassification.from_pretrained('distilber
```

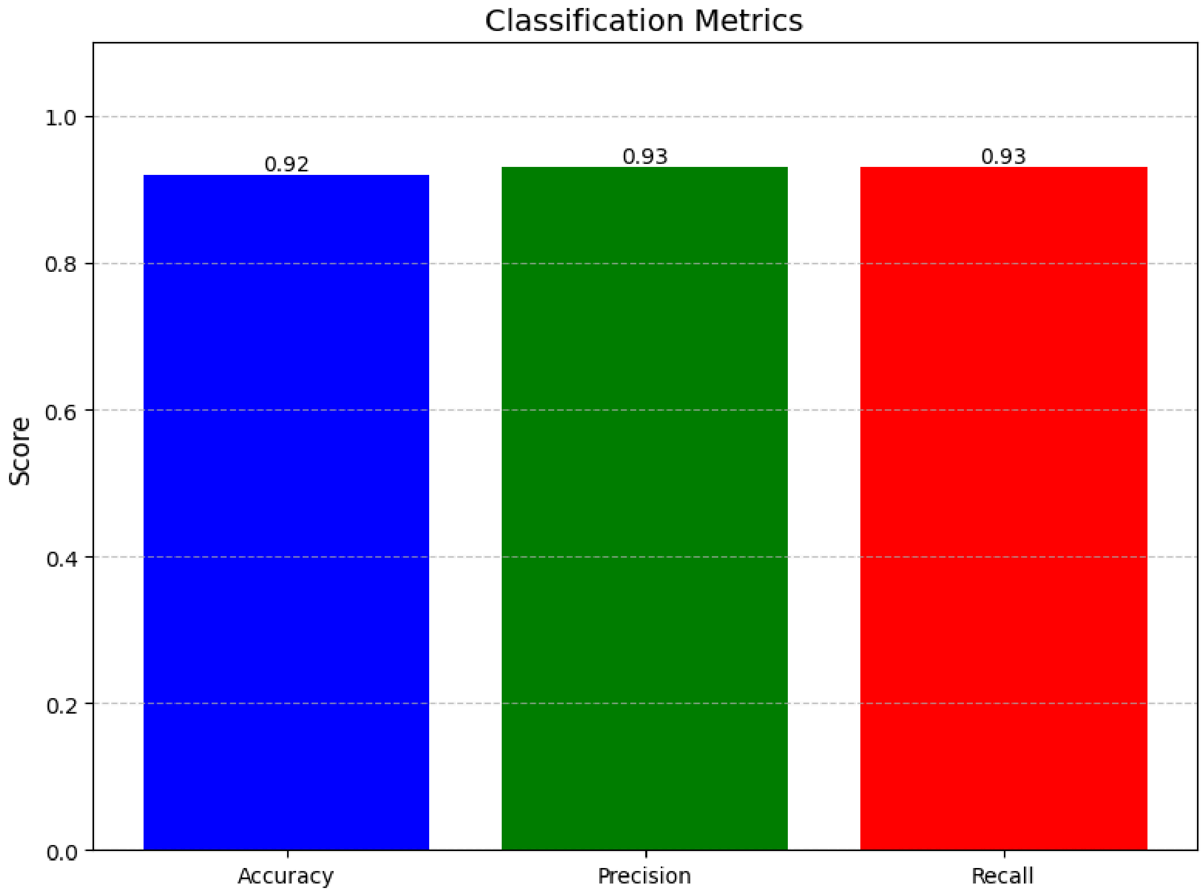
```
model_hotel.load_state_dict(checkpoint["model_state_dict"])

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model_hotel.to(device)

test_true, test_preds = evaluate2(model_hotel, device, test_loader_hotel)
accuracy = accuracy_score(test_true, test_preds)
print(f"Test Accuracy: {accuracy * 100:.2f}%\n")
```

```
Some weights of DistilBertForSequenceClassification were not initialized fro
m the model checkpoint at distilbert-base-uncased and are newly initialized:
['classifier.bias', 'classifier.weight', 'pre_classifier.bias', 'pre_classif
ier.weight']
You should probably TRAIN this model on a down-stream task to be able to use
it for predictions and inference.
Evaluating: 100%|██████████| 229/229 [00:08<00:00, 28.45it/s]
Test Accuracy: 92.90%
```

In [30]:
```
plot_confusion_matrix(test_true, test_preds, labels=["Negative", "Positive"]
```

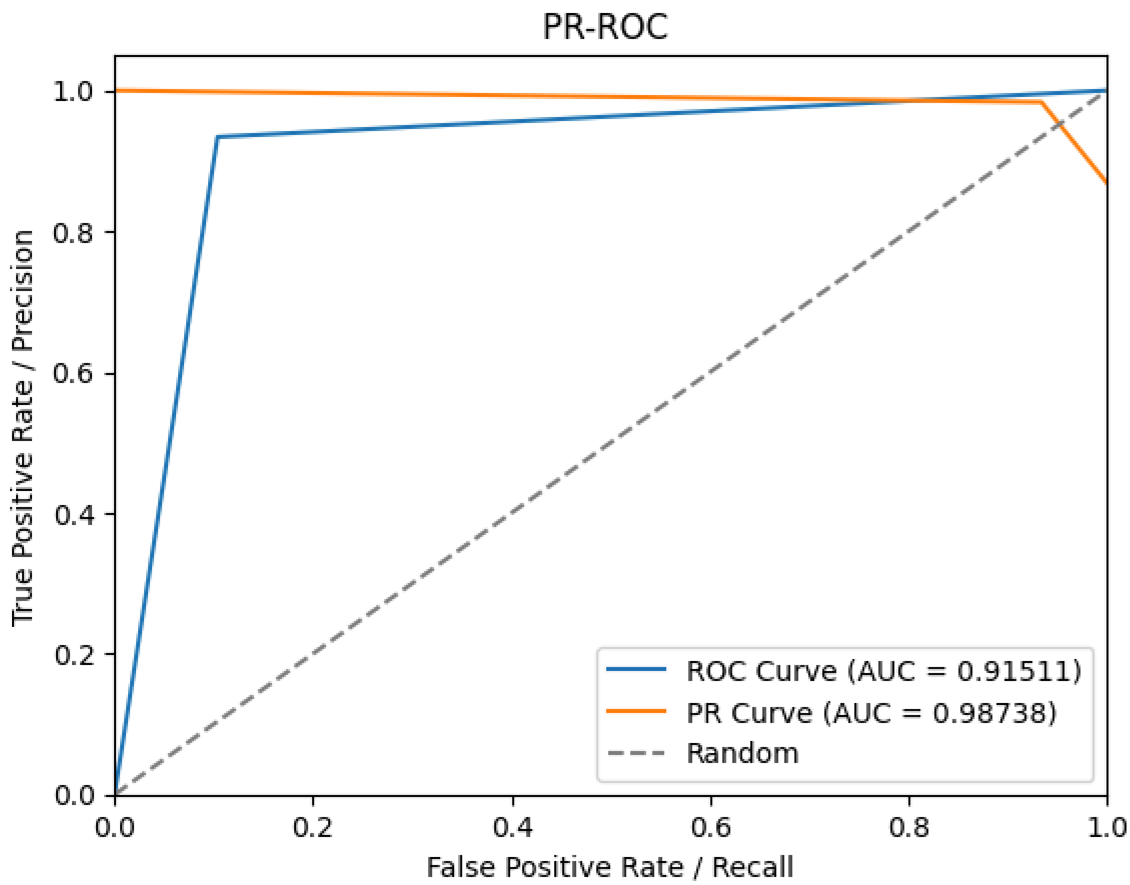


In [126…



In [124…

```
plot_auc(test_preds, test_true)
```

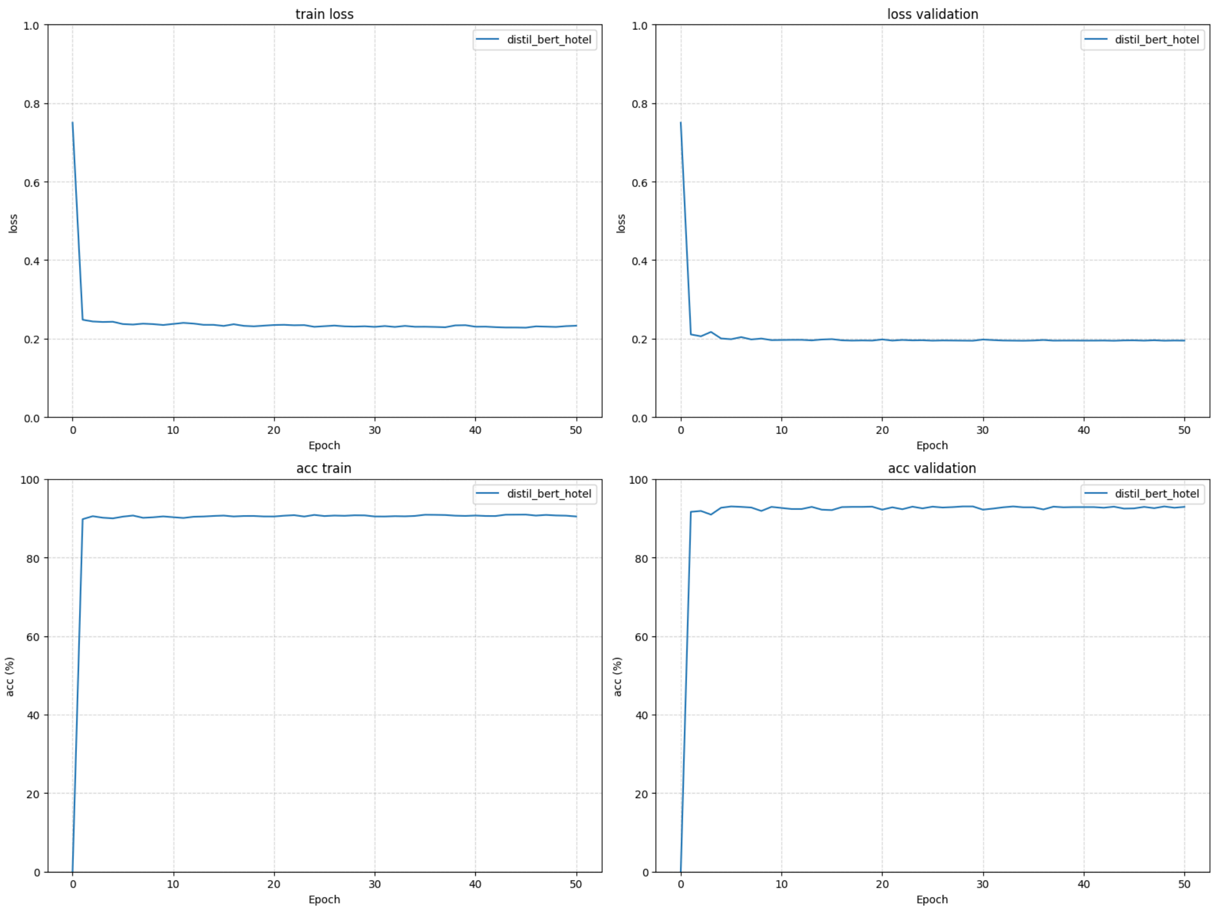


In [ ]:
```
def plot_training_curves(history, model_names, save_path='training_curves_cc
    plt.figure(figsize=(16, 12))

    plt.subplot(2, 2, 1)
    plt.plot([1] + history['train_losses'], label=f'{model_names}')

    plt.xlabel('Epoch')
    plt.ylabel('loss ')
    plt.title('loss acc')
    plt.ylim(0.0, 1)
    plt.legend()
    plt.grid(linestyle='--', alpha=0.5)

    plt.subplot(2, 2, 2)
    plt.plot([0] + history['train_accuracies'], label=f'{model_names}')
    plt.xlabel('Epoch')
    plt.ylabel('acc (%)')
    plt.ylim(0.0, 100)
    plt.title('loss acc')
    plt.legend()
    plt.grid(linestyle='--', alpha=0.5)

    plt.tight_layout()
    plt.show()

def plot_training_curves2(history, model_names, save_path='training_curves_c
    plt.figure(figsize=(16, 12))
    # set start from 0
```

```
    history['train_losses'] = np.insert(history['train_losses'], 0, 0.75)
    history['val_losses'] = np.insert(history['val_losses'], 0, 0.75)
    history['train_accuracies'] = np.insert(history['train_accuracies'], 0,
    history['val_accuracies'] = np.insert(history['val_accuracies'], 0, 0)

    # plot train loss curve
    plt.subplot(2, 2, 1)
    plt.plot(history['train_losses'], label=f'{model_names}')
    plt.xlabel('Epoch')
    plt.ylabel('loss')
    plt.title('train loss')
    plt.ylim(0.0, 1)
    plt.legend()
    plt.grid(linestyle='--', alpha=0.5)

    # plot validation loss curve
    plt.subplot(2, 2, 2)
    plt.plot(history['val_losses'], label=f'{model_names}')
    plt.xlabel('Epoch')
    plt.ylabel('loss')
    plt.title('loss validation')
    plt.ylim(0.0, 1)
    plt.legend()
    plt.grid(linestyle='--', alpha=0.5)

    # plot train accuracy curve
    plt.subplot(2, 2, 3)
    plt.plot(history['train_accuracies'], label=f'{model_names}')
    plt.xlabel('Epoch')
    plt.ylabel('acc (%)')
    plt.title('acc train')
    plt.ylim(0.0, 100)
    plt.legend()
    plt.grid(linestyle='--', alpha=0.5)

    # plot validation accuracy curve
    plt.subplot(2, 2, 4)
    plt.plot(history['val_accuracies'], label=f'{model_names}')
    plt.xlabel('Epoch')
    plt.ylabel('acc (%)')
    plt.title('acc validation')
    plt.ylim(0.0, 100)

    plt.legend()
    plt.grid(linestyle='--', alpha=0.5)

    plt.tight_layout()
    plt.show()
```

In [123…

```
plot_training_curves2(history_hotel, 'distil_bert_hotel')
```



In [48]:

```
from types import SimpleNamespace
class TextCNN(nn.Module):
    def __init__(self, vocab_size, embed_dim, num_classes, kernel_sizes=[3],
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.convs = nn.ModuleList([
            nn.Conv2d(1, num_filters, (k, embed_dim)) for k in kernel_sizes
        ])
        self.dropout = nn.Dropout(0.7)
        self.fc = nn.Linear(num_filters * len(kernel_sizes), num_classes)

    def forward(self, input_ids, attention_mask=None,  labels=None):
        x = self.embedding(input_ids)  # [B, L, D]
        x = x.unsqueeze(1)             # [B, 1, L, D]
        x = [torch.relu(conv(x)).squeeze(3) for conv in self.convs]
        x = [torch.max(p, dim=2)[0] for p in x]
        x = torch.cat(x, dim=1)
        x = self.dropout(x)
        logits = self.fc(x)

        if labels is not None:
            loss = nn.CrossEntropyLoss()(logits, labels)
            return SimpleNamespace(loss=loss, logits=logits)
        else:
            return SimpleNamespace(logits=logits)

        # return self.fc(x)
```

In [32]:
```
class SentimentDataset2(Dataset):
    def __init__(self, texts, labels, tokenizer, max_len=64):
        self.texts = texts
        self.labels = labels
        self.tokenizer = tokenizer
        self.max_len = max_len

    def __len__(self):
        return len(self.texts)

    def __getitem__(self, idx):
        encoding = self.tokenizer.encode_plus(
            self.texts[idx],
            add_special_tokens=True,
            max_length=self.max_len,
            padding='max_length',
            truncation=True,
            return_tensors='pt'
        )
        return {
            'input_ids': encoding['input_ids'].squeeze(0),  # [L]
            'attention_mask': encoding['attention_mask'].squeeze(0),
            'labels': torch.tensor(self.labels[idx], dtype=torch.long)
        }
```

TextCNN

In [ ]:
```
# ========== data processing ==========

data_collator = DataCollatorWithPadding(tokenizer_hotel)
train_dataset_cnn = SentimentDataset2(train_df_hotel["processed_text"].tolis
                                      train_df_hotel['sentiment_encode'].tol
test_dataset_cnn = SentimentDataset2(test_df_hotel["processed_text"].tolist(
                                     test_df_hotel['sentiment_encode'].tolis
train_loader_cnn = DataLoader(train_dataset_cnn, batch_size=16, shuffle=True
test_loader_cnn = DataLoader(test_dataset_cnn, batch_size=16, shuffle=False,


vocab_size = tokenizer_hotel.vocab_size  #  embedding initialization
cnn_model = TextCNN(vocab_size, embed_dim=128, num_classes=2)


history_hotel_cnn = train(cnn_model,
      train_loader_cnn,
      test_loader_cnn,
      save_path="distil_cnn_hotel.pth",
      # fine_tuning = True,
      # target_names = ['negative', 'neutral', 'positive'],
      target_names = ['negative', 'positive'],
      num_epochs=50
     )
```

```
训练中 1: 100%|██████████| 986/986 [00:12<00:00, 79.40it/s, loss=0.559, acc=7
5.8]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 117.20it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.00      0.00      0.00       302
    positive       0.83      1.00      0.91      1450

    accuracy                           0.83      1752
   macro avg       0.41      0.50      0.45      1752
weighted avg       0.68      0.83      0.75      1752

训练中 2: 100%|██████████| 986/986 [00:12<00:00, 80.85it/s, loss=0.433, acc=8
2.9]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.60it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       1.00      0.01      0.02       302
    positive       0.83      1.00      0.91      1450

    accuracy                           0.83      1752
   macro avg       0.91      0.50      0.46      1752
weighted avg       0.86      0.83      0.75      1752

训练中 3: 100%|██████████| 986/986 [00:12<00:00, 80.87it/s, loss=0.397, acc=8
4.1]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.90it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.97      0.12      0.22       302
    positive       0.85      1.00      0.92      1450

    accuracy                           0.85      1752
   macro avg       0.91      0.56      0.57      1752
weighted avg       0.87      0.85      0.80      1752

训练中 4: 100%|██████████| 986/986 [00:12<00:00, 80.72it/s, loss=0.365, acc=8
5.8]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 118.78it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.95      0.23      0.37       302
    positive       0.86      1.00      0.92      1450

    accuracy                           0.86      1752
   macro avg       0.90      0.61      0.65      1752
weighted avg       0.88      0.86      0.83      1752

训练中 5: 100%|██████████| 986/986 [00:12<00:00, 80.35it/s, loss=0.345, acc=8
6.4]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 118.46it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.95      0.33      0.49       302
    positive       0.88      1.00      0.93      1450

    accuracy                           0.88      1752
   macro avg       0.91      0.66      0.71      1752
weighted avg       0.89      0.88      0.86      1752
```

```
训练中 6: 100%|██████████| 986/986 [00:12<00:00, 80.62it/s, loss=0.321, acc=8
7.5]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 116.41it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.94      0.39      0.56       302
    positive       0.89      1.00      0.94      1450

    accuracy                           0.89      1752
   macro avg       0.92      0.69      0.75      1752
weighted avg       0.90      0.89      0.87      1752
```

```
训练中 7: 100%|██████████| 986/986 [00:12<00:00, 80.59it/s, loss=0.301, acc=8
8.3]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.63it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.46      0.61       302
    positive       0.90      0.99      0.94      1450

    accuracy                           0.90      1752
   macro avg       0.90      0.72      0.78      1752
weighted avg       0.90      0.90      0.88      1752
```

```
训练中 8: 100%|██████████| 986/986 [00:12<00:00, 80.64it/s, loss=0.289, acc=8
8.9]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 118.55it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.91      0.56      0.69       302
    positive       0.92      0.99      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.91      0.77      0.82      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 9: 100%|██████████| 986/986 [00:12<00:00, 79.63it/s, loss=0.281, acc=8
9.4]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 117.91it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.50      0.64       302
    positive       0.90      0.99      0.94      1450

    accuracy                           0.90      1752
   macro avg       0.90      0.74      0.79      1752
weighted avg       0.90      0.90      0.89      1752

训练中 10: 100%|██████████| 986/986 [00:12<00:00, 81.06it/s, loss=0.27, acc=8
9.7]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.85it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.56      0.69       302
    positive       0.91      0.99      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.91      0.77      0.82      1752
weighted avg       0.91      0.91      0.90      1752

训练中 11: 100%|██████████| 986/986 [00:12<00:00, 80.75it/s, loss=0.264, acc=
89.9]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.86it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.93      0.50      0.65       302
    positive       0.91      0.99      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.92      0.75      0.80      1752
weighted avg       0.91      0.91      0.90      1752

训练中 12: 100%|██████████| 986/986 [00:12<00:00, 80.95it/s, loss=0.252, acc=
90.4]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 122.61it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.88      0.58      0.70       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.90      0.78      0.83      1752
weighted avg       0.91      0.91      0.91      1752

训练中 13: 100%|██████████| 986/986 [00:12<00:00, 81.48it/s, loss=0.247, acc=
90.7]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 121.21it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.59      0.71       302
    positive       0.92      0.99      0.95      1450

    accuracy                           0.92      1752
   macro avg       0.91      0.79      0.83      1752
weighted avg       0.92      0.92      0.91      1752
```

```
训练中 14: 100%|██████████| 986/986 [00:12<00:00, 81.30it/s, loss=0.238, acc=
90.9]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 118.21it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.87      0.62      0.72       302
    positive       0.93      0.98      0.95      1450

    accuracy                           0.92      1752
   macro avg       0.90      0.80      0.84      1752
weighted avg       0.92      0.92      0.91      1752
```

```
训练中 15: 100%|██████████| 986/986 [00:12<00:00, 80.62it/s, loss=0.229, acc=
91.2]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.08it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.60      0.71       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.79      0.83      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 16: 100%|██████████| 986/986 [00:12<00:00, 81.15it/s, loss=0.227, acc=
91.1]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.47it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.86      0.58      0.70       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.78      0.82      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 17: 100%|██████████| 986/986 [00:12<00:00, 80.81it/s, loss=0.222, acc=
91.3]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 122.41it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.87      0.57      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.90      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 18: 100%|██████████| 986/986 [00:12<00:00, 81.03it/s, loss=0.217, acc=
91.7]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 123.33it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.90      0.54      0.67       302
    positive       0.91      0.99      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.90      0.76      0.81      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 19: 100%|██████████| 986/986 [00:12<00:00, 81.36it/s, loss=0.209, acc=
91.6]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 120.71it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.63      0.72       302
    positive       0.93      0.98      0.95      1450

    accuracy                           0.92      1752
   macro avg       0.89      0.80      0.84      1752
weighted avg       0.91      0.92      0.91      1752
```

```
训练中 20: 100%|██████████| 986/986 [00:12<00:00, 81.20it/s, loss=0.202, acc=
92]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 120.57it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.83      0.61      0.70       302
    positive       0.92      0.97      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.87      0.79      0.83      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 21: 100%|██████████| 986/986 [00:12<00:00, 80.12it/s, loss=0.202, acc=
92]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 118.46it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.84      0.61      0.71       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.79      0.83      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 22: 100%|██████████| 986/986 [00:12<00:00, 81.07it/s, loss=0.201, acc=
92.1]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 123.46it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.57      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 23: 100%|██████████| 986/986 [00:11<00:00, 83.47it/s, loss=0.2, acc=9
2.1]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 125.49it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.86      0.58      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 24: 100%|██████████| 986/986 [00:12<00:00, 80.85it/s, loss=0.194, acc=
92.2]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 120.10it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.88      0.60      0.71       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.92      1752
   macro avg       0.90      0.79      0.83      1752
weighted avg       0.92      0.92      0.91      1752
```

```
训练中 25: 100%|██████████| 986/986 [00:12<00:00, 80.14it/s, loss=0.185, acc=
92.7]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.54it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.61      0.71       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.79      0.83      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 26: 100%|██████████| 986/986 [00:12<00:00, 79.90it/s, loss=0.184, acc=
92.5]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 116.31it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.84      0.58      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 27: 100%|██████████| 986/986 [00:12<00:00, 80.96it/s, loss=0.192, acc=
92.4]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 120.35it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.83      0.59      0.69       302
    positive       0.92      0.97      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.87      0.78      0.82      1752
weighted avg       0.90      0.91      0.90      1752
```

```
训练中 28: 100%|██████████| 986/986 [00:12<00:00, 79.82it/s, loss=0.185, acc=
92.4]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 117.48it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.84      0.62      0.71       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.80      0.83      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 29: 100%|██████████| 986/986 [00:12<00:00, 81.29it/s, loss=0.182, acc=
92.7]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 117.93it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.58      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752

训练中 30: 100%|██████████| 986/986 [00:12<00:00, 80.68it/s, loss=0.181, acc=
92.8]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.01it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.56      0.68       302
    positive       0.91      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.77      0.81      1752
weighted avg       0.90      0.91      0.90      1752

训练中 31: 100%|██████████| 986/986 [00:12<00:00, 80.21it/s, loss=0.18, acc=9
2.6]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.68it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.57      0.68       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.77      0.82      1752
weighted avg       0.91      0.91      0.90      1752

训练中 32: 100%|██████████| 986/986 [00:12<00:00, 81.35it/s, loss=0.174, acc=
92.8]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 122.11it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.57      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752

训练中 33: 100%|██████████| 986/986 [00:11<00:00, 83.73it/s, loss=0.178, acc=
92.8]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.81it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.88      0.57      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.90      0.77      0.82      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 34: 100%|██████████| 986/986 [00:12<00:00, 82.04it/s, loss=0.18, acc=9
2.7]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.77it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.59      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 35: 100%|██████████| 986/986 [00:12<00:00, 81.15it/s, loss=0.175, acc=
92.7]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 126.69it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.86      0.61      0.71       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.92      1752
   macro avg       0.89      0.80      0.83      1752
weighted avg       0.91      0.92      0.91      1752
```

```
训练中 36: 100%|██████████| 986/986 [00:11<00:00, 83.77it/s, loss=0.171, acc=
92.7]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 132.16it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.59      0.70       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.79      0.82      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 37: 100%|██████████| 986/986 [00:11<00:00, 85.58it/s, loss=0.162, acc=
93.1]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 127.62it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.59      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 38: 100%|██████████| 986/986 [00:11<00:00, 83.19it/s, loss=0.161, acc=93.2]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 124.36it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.83      0.58      0.68       302
    positive       0.92      0.97      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.87      0.78      0.81      1752
weighted avg       0.90      0.91      0.90      1752
```

```
训练中 39: 100%|██████████| 986/986 [00:12<00:00, 80.90it/s, loss=0.163, acc=93]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 118.78it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.84      0.57      0.68       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.77      0.81      1752
weighted avg       0.90      0.91      0.90      1752
```

```
训练中 40: 100%|██████████| 986/986 [00:12<00:00, 80.51it/s, loss=0.158, acc=93.3]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 117.92it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.57      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 41: 100%|██████████| 986/986 [00:12<00:00, 79.94it/s, loss=0.163, acc=93.1]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 120.69it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.60      0.70       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.79      0.83      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 42: 100%|██████████| 986/986 [00:12<00:00, 81.40it/s, loss=0.16, acc=9
3.2]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 119.22it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.83      0.58      0.68       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.87      0.78      0.81      1752
weighted avg       0.90      0.91      0.90      1752
```

```
训练中 43: 100%|██████████| 986/986 [00:12<00:00, 81.59it/s, loss=0.159, acc=
93.1]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 117.19it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.84      0.57      0.68       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.77      0.81      1752
weighted avg       0.90      0.91      0.90      1752
```

```
训练中 44: 100%|██████████| 986/986 [00:12<00:00, 81.26it/s, loss=0.156, acc=
93.2]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 122.43it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.58      0.69       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.78      0.82      1752
weighted avg       0.91      0.91      0.90      1752
```

```
训练中 45: 100%|██████████| 986/986 [00:12<00:00, 81.20it/s, loss=0.154, acc=
93.4]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 117.62it/s]
```

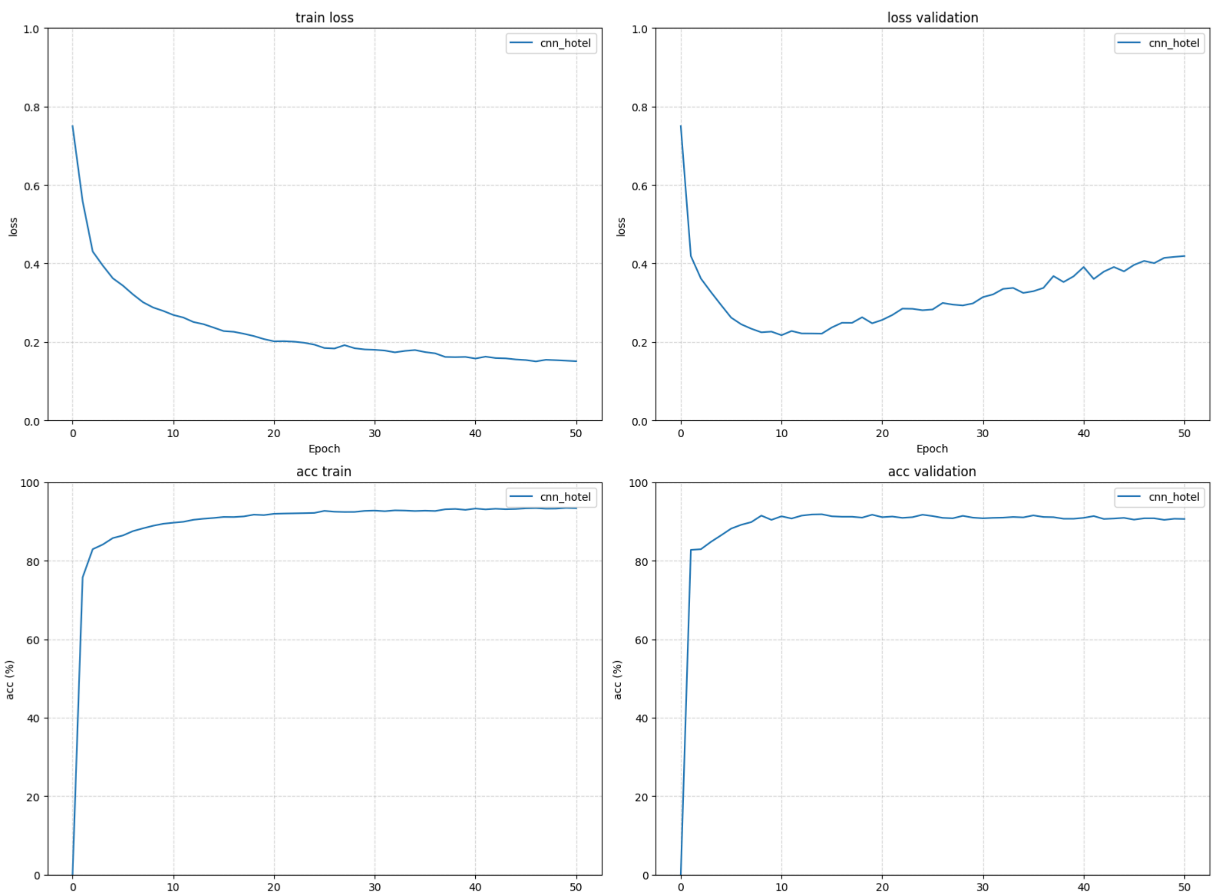
```
Classification Report:
               precision    recall  f1-score   support

    negative       0.84      0.56      0.67       302
    positive       0.91      0.98      0.94      1450

    accuracy                           0.90      1752
   macro avg       0.87      0.77      0.81      1752
weighted avg       0.90      0.90      0.90      1752

训练中 46: 100%|██████████| 986/986 [00:12<00:00, 80.73it/s, loss=0.15, acc=93.4]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 120.30it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.56      0.68       302
    positive       0.91      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.89      0.77      0.81      1752
weighted avg       0.90      0.91      0.90      1752

训练中 47: 100%|██████████| 986/986 [00:12<00:00, 80.61it/s, loss=0.155, acc=93.3]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 116.84it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.57      0.68       302
    positive       0.92      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.77      0.81      1752
weighted avg       0.90      0.91      0.90      1752

训练中 48: 100%|██████████| 986/986 [00:12<00:00, 81.22it/s, loss=0.154, acc=93.3]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 121.71it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.54      0.66       302
    positive       0.91      0.98      0.94      1450

    accuracy                           0.90      1752
   macro avg       0.88      0.76      0.80      1752
weighted avg       0.90      0.90      0.90      1752

训练中 49: 100%|██████████| 986/986 [00:12<00:00, 81.79it/s, loss=0.154, acc=93.5]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 123.25it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.56      0.67       302
    positive       0.91      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.77      0.81      1752
weighted avg       0.90      0.91      0.90      1752
```

```
训练中 50: 100%|██████████| 986/986 [00:12<00:00, 81.84it/s, loss=0.152, acc=
93.4]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 121.48it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.84      0.56      0.67       302
    positive       0.91      0.98      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.88      0.77      0.81      1752
weighted avg       0.90      0.91      0.90      1752


Saving model...distil_cnn_hotel.pth
```

In [50]:
```
plot_training_curves2(history_hotel_cnn, 'cnn_hotel')
```



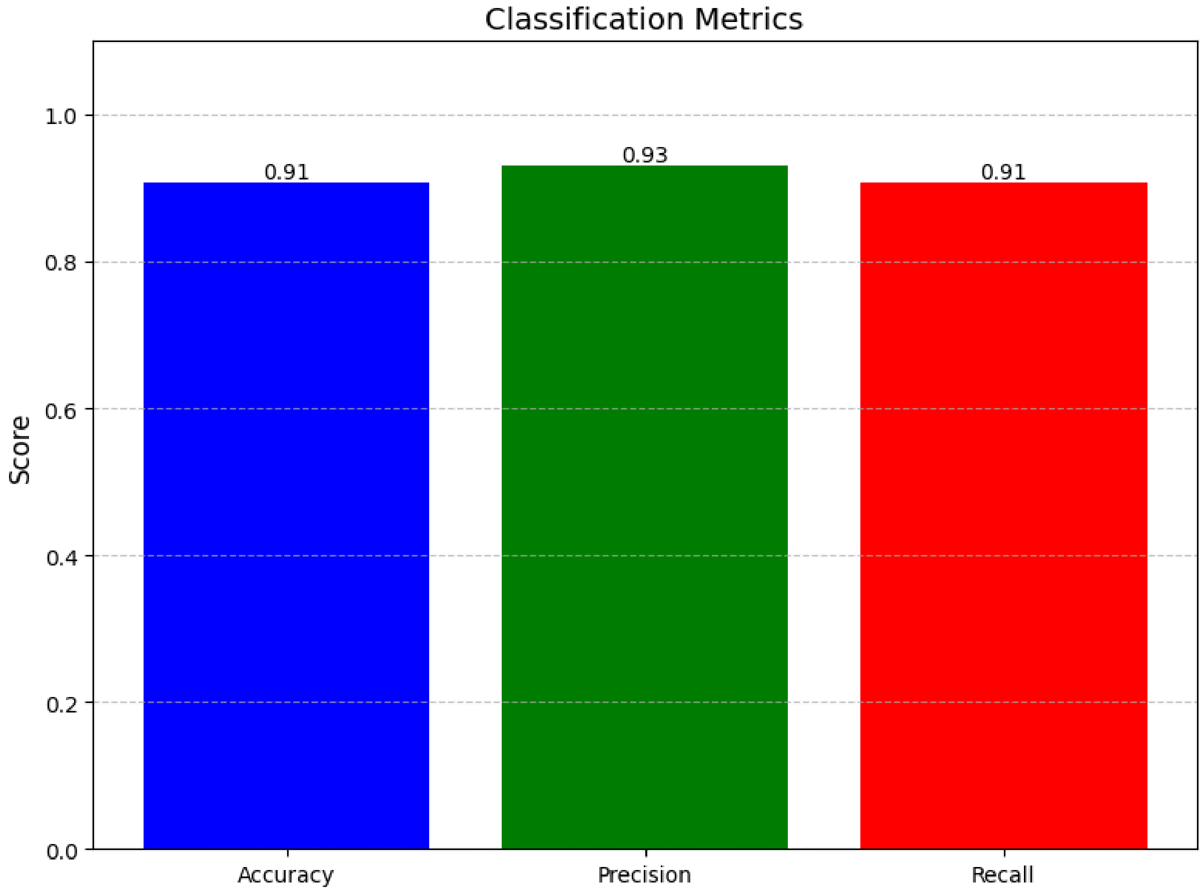
In [80]:
```
cnn_model
```

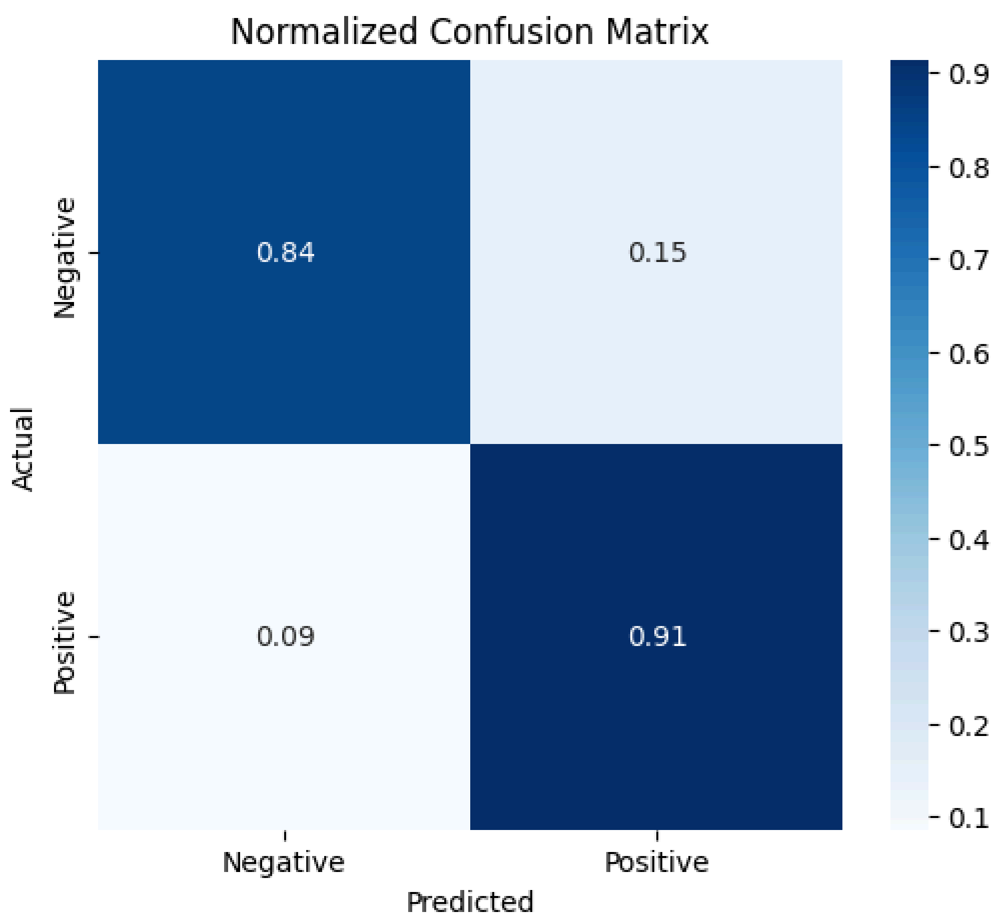
Out[80]:
```
TextCNN(
  (embedding): Embedding(30522, 128, padding_idx=0)
  (convs): ModuleList(
    (0): Conv2d(1, 8, kernel_size=(3, 128), stride=(1, 1))
  )
  (dropout): Dropout(p=0.7, inplace=False)
  (fc): Linear(in_features=8, out_features=2, bias=True)
)
```

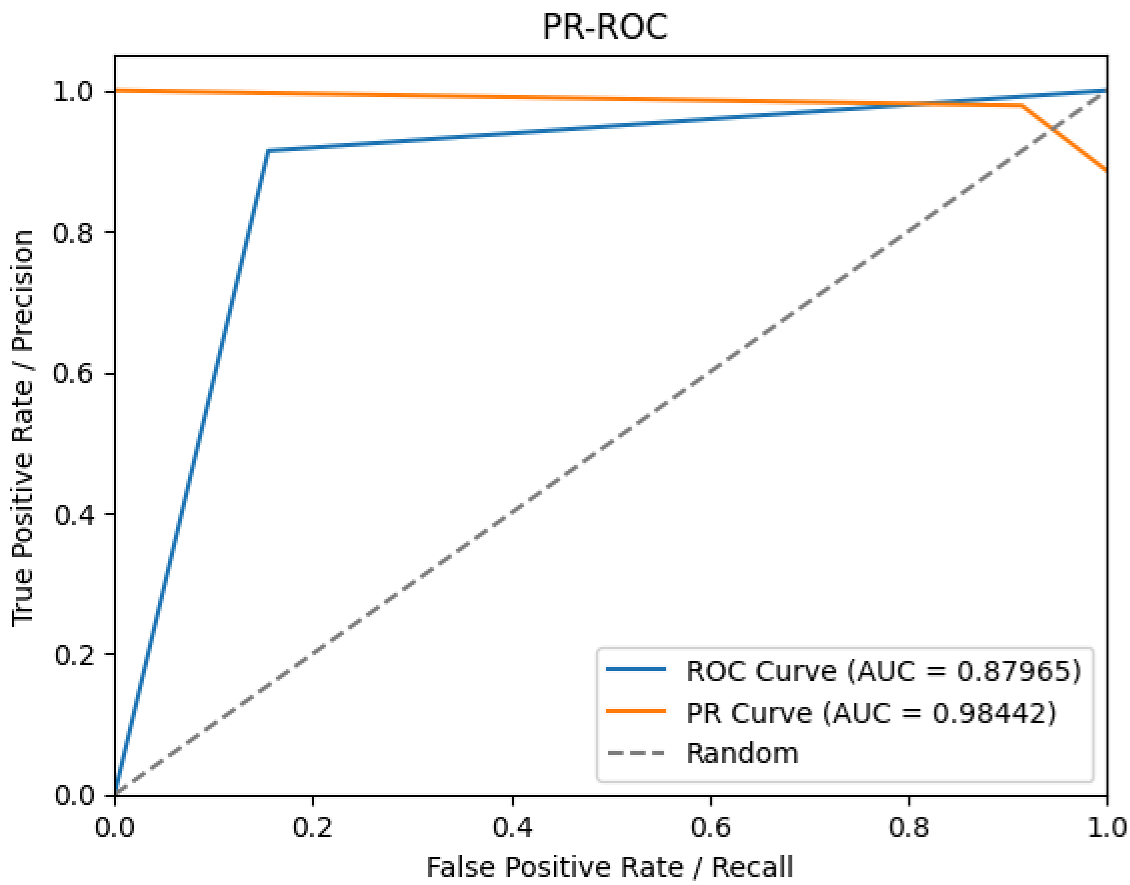
In [76]:
```
# checkpoint = torch.load("distil_cnn_hotel.pth")
# model_hotel = DistilBertForSequenceClassification.from_pretrained('distilb
# model_hotel.load_state_dict(checkpoint["model_state_dict"])

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
cnn_model.to(device)
# all_preds, all_labels, avg_loss, acc*100.
test_true, test_preds, _,_ = evaluate2(cnn_model, device, test_loader_cnn)
accuracy = accuracy_score(test_true, test_preds)
print(f"Test Accuracy: {accuracy * 100:.2f}%\n")
```

```
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 105.21it/s]
Test Accuracy: 90.64%
```

In [77]:
```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f

acc = accuracy_score(test_true, test_preds)
precision = precision_score(test_true, test_preds, average='weighted')  # 或
recall = recall_score(test_true, test_preds, average='weighted')
f1 = f1_score(test_true, test_preds, average='weighted')
draw_bar(acc, precision, recall)
plot_confusion_matrix(test_true, test_preds, labels=["Negative", "Positive"]
plot_auc(test_preds, test_true)
```


Classification Metrics
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.91
0.93
0.91
Accuracy
Precision
Recall


Normalized Confusion Matrix
Actual
Negative
Positive
0.84
0.15
0.09
0.91
Negative
Positive
Predicted
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1



LSTM

In [59]:
```
class Attention(nn.Module):
    def __init__(self, hidden_dim):
        super().__init__()
        self.attn = nn.Linear(hidden_dim * 2, 1)

    def forward(self, lstm_output):
        weights = torch.softmax(self.attn(lstm_output), dim=1)
        context = torch.sum(weights * lstm_output, dim=1)
        return context

class BiLSTM_Attn(nn.Module):
    def __init__(self, vocab_size, embed_dim, hidden_dim, num_classes):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.lstm = nn.LSTM(embed_dim, hidden_dim, batch_first=True, bidirec
        # self.attn = Attention(hidden_dim)
        self.attn = None
        self.fc = nn.Linear(hidden_dim * 2, num_classes)
        self.dropout = nn.Dropout(0.7)

    def forward(self, input_ids, attention_mask=None, labels=None):
        x = self.embedding(input_ids)
        lstm_out, _ = self.lstm(x)
```

```
        if self.attn:
            x = self.attn(lstm_out)
        else:
            x = lstm_out[:, -1, :]

        x = self.dropout(x)
        logits = self.fc(x)

        if labels is not None:
            loss = nn.CrossEntropyLoss()(logits, labels)
            return SimpleNamespace(loss=loss, logits=logits)
        else:
            return SimpleNamespace(logits=logits)
```

In [60]:

```
model_lstm = BiLSTM_Attn(vocab_size, embed_dim=32, hidden_dim=32, num_classe

history_hotel_lstm = train(model_lstm,
      train_loader_cnn,
      test_loader_cnn,
      save_path="distil_lstm_hotel.pth",
      # fine_tuning = True,
      # target_names = ['negative', 'neutral', 'positive'],
      target_names = ['negative', 'positive'],
      num_epochs=50
     )
```

```
训练中 1:   0%|          | 0/986 [00:00<?, ?it/s]
训练中 1: 100%|██████████| 986/986 [00:12<00:00, 76.44it/s, loss=0.427, acc=8
2.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.39it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.73      0.39      0.51       302
    positive       0.88      0.97      0.92      1450

    accuracy                           0.87      1752
   macro avg       0.81      0.68      0.72      1752
weighted avg       0.86      0.87      0.85      1752

训练中 2: 100%|██████████| 986/986 [00:12<00:00, 76.30it/s, loss=0.418, acc=8
3.8]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.98it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.78      0.45      0.57       302
    positive       0.89      0.97      0.93      1450

    accuracy                           0.88      1752
   macro avg       0.84      0.71      0.75      1752
weighted avg       0.87      0.88      0.87      1752
```

```
训练中 3: 100%|██████████| 986/986 [00:12<00:00, 76.06it/s, loss=0.309, acc=8
8.6]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.42it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.66      0.74      0.70       302
    positive       0.94      0.92      0.93      1450

    accuracy                           0.89      1752
   macro avg       0.80      0.83      0.82      1752
weighted avg       0.90      0.89      0.89      1752
```

```
训练中 4: 100%|██████████| 986/986 [00:12<00:00, 75.98it/s, loss=0.318, acc=8
8.7]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.21it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.60      0.49      0.54       302
    positive       0.90      0.93      0.91      1450

    accuracy                           0.86      1752
   macro avg       0.75      0.71      0.73      1752
weighted avg       0.85      0.86      0.85      1752
```

```
训练中 5: 100%|██████████| 986/986 [00:12<00:00, 75.93it/s, loss=0.271, acc=8
9.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.16it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.78      0.69      0.73       302
    positive       0.94      0.96      0.95      1450

    accuracy                           0.91      1752
   macro avg       0.86      0.82      0.84      1752
weighted avg       0.91      0.91      0.91      1752
```

```
训练中 6: 100%|██████████| 986/986 [00:12<00:00, 76.00it/s, loss=0.218, acc=9
2.1]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.79it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.76      0.77      0.77       302
    positive       0.95      0.95      0.95      1450

    accuracy                           0.92      1752
   macro avg       0.86      0.86      0.86      1752
weighted avg       0.92      0.92      0.92      1752
```

```
训练中 7: 100%|██████████| 986/986 [00:12<00:00, 76.18it/s, loss=0.175, acc=9
3.8]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.71it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.83      0.76      0.80       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.86      0.88      1752
weighted avg       0.93      0.93      0.93      1752

训练中 8: 100%|██████████| 986/986 [00:12<00:00, 76.49it/s, loss=0.158, acc=9
4.7]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.37it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.87      0.70      0.78       302
    positive       0.94      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.91      0.84      0.87      1752
weighted avg       0.93      0.93      0.93      1752

训练中 9: 100%|██████████| 986/986 [00:12<00:00, 76.58it/s, loss=0.125, acc=9
6.2]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.90it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.72      0.78       302
    positive       0.94      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752

训练中 10: 100%|██████████| 986/986 [00:13<00:00, 75.73it/s, loss=0.105, acc=
97.1]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.38it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.75      0.80       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.86      0.88      1752
weighted avg       0.93      0.93      0.93      1752

训练中 11: 100%|██████████| 986/986 [00:12<00:00, 76.52it/s, loss=0.0841, acc
=97.6]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.52it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.80      0.80      0.80       302
    positive       0.96      0.96      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.88      0.88      0.88      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 12: 100%|██████████| 986/986 [00:12<00:00, 75.98it/s, loss=0.0716, acc
=98.1]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 106.86it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.83      0.77      0.80       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.87      0.88      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 13: 100%|██████████| 986/986 [00:12<00:00, 76.08it/s, loss=0.0599, acc
=98.5]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.09it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.89      0.68      0.77       302
    positive       0.94      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.91      0.83      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 14: 100%|██████████| 986/986 [00:12<00:00, 76.24it/s, loss=0.0493, acc
=98.6]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.15it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.75      0.78      0.77       302
    positive       0.95      0.95      0.95      1450

    accuracy                           0.92      1752
   macro avg       0.85      0.87      0.86      1752
weighted avg       0.92      0.92      0.92      1752
```

```
训练中 15: 100%|██████████| 986/986 [00:13<00:00, 75.30it/s, loss=0.0362, acc
=99.1]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.44it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.86      0.75      0.80       302
    positive       0.95      0.98      0.96      1450

    accuracy                           0.94      1752
   macro avg       0.91      0.86      0.88      1752
weighted avg       0.93      0.94      0.93      1752
```

```
训练中 16: 100%|██████████| 986/986 [00:13<00:00, 75.39it/s, loss=0.0393, acc
=99]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.54it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.83      0.78      0.81       302
    positive       0.96      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.88      0.88      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 17: 100%|██████████| 986/986 [00:13<00:00, 75.42it/s, loss=0.0286, acc
=99.3]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.11it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.84      0.74      0.78       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 18: 100%|██████████| 986/986 [00:12<00:00, 76.46it/s, loss=0.0309, acc
=99.3]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.71it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.78      0.78      0.78       302
    positive       0.95      0.96      0.95      1450

    accuracy                           0.93      1752
   macro avg       0.87      0.87      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 19: 100%|██████████| 986/986 [00:12<00:00, 76.37it/s, loss=0.0236, acc
=99.4]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.27it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.83      0.77      0.80       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.87      0.88      1752
weighted avg       0.93      0.93      0.93      1752

训练中 20: 100%|██████████| 986/986 [00:12<00:00, 76.36it/s, loss=0.022, acc=
99.5]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 101.29it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.83      0.75      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.86      0.87      1752
weighted avg       0.93      0.93      0.93      1752

训练中 21: 100%|██████████| 986/986 [00:12<00:00, 75.94it/s, loss=0.0197, acc
=99.5]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.83it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.82      0.75      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.86      0.87      1752
weighted avg       0.93      0.93      0.93      1752

训练中 22: 100%|██████████| 986/986 [00:12<00:00, 76.36it/s, loss=0.0154, acc
=99.7]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 105.80it/s]
Classification Report:
               precision    recall  f1-score   support

    negative       0.88      0.70      0.78       302
    positive       0.94      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.91      0.84      0.87      1752
weighted avg       0.93      0.93      0.93      1752

训练中 23: 100%|██████████| 986/986 [00:12<00:00, 76.40it/s, loss=0.0152, acc
=99.7]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 105.03it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.82      0.77      0.80       302
    positive       0.95      0.96      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.87      0.88      1752
weighted avg       0.93      0.93      0.93      1752

训练中 24: 100%|██████████| 986/986 [00:12<00:00, 76.39it/s, loss=0.0146, acc
=99.7]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.79it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.83      0.74      0.78       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752

训练中 25: 100%|██████████| 986/986 [00:12<00:00, 76.87it/s, loss=0.0107, acc
=99.8]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.26it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.74      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.86      0.88      1752
weighted avg       0.93      0.93      0.93      1752

训练中 26: 100%|██████████| 986/986 [00:12<00:00, 77.09it/s, loss=0.0126, acc
=99.8]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 105.48it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.84      0.68      0.75       302
    positive       0.94      0.97      0.95      1450

    accuracy                           0.92      1752
   macro avg       0.89      0.83      0.85      1752
weighted avg       0.92      0.92      0.92      1752

训练中 27: 100%|██████████| 986/986 [00:12<00:00, 76.85it/s, loss=0.0193, acc
=99.5]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 105.60it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.79      0.77      0.78       302
    positive       0.95      0.96      0.95      1450

    accuracy                           0.92      1752
   macro avg       0.87      0.86      0.87      1752
weighted avg       0.92      0.92      0.92      1752
```

```
训练中 28: 100%|██████████| 986/986 [00:12<00:00, 77.07it/s, loss=0.0129, acc
=99.7]
Evaluating: 100%|██████████| 110/110 [00:00<00:00, 110.93it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.70      0.77       302
    positive       0.94      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.84      0.86      1752
weighted avg       0.92      0.93      0.92      1752
```

```
训练中 29: 100%|██████████| 986/986 [00:12<00:00, 76.63it/s, loss=0.00751, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.23it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.82      0.73      0.78       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.85      0.87      1752
weighted avg       0.92      0.93      0.93      1752
```

```
训练中 30: 100%|██████████| 986/986 [00:12<00:00, 76.10it/s, loss=0.00634, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.57it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.82      0.75      0.78       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.88      0.86      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 31: 100%|██████████| 986/986 [00:12<00:00, 76.29it/s, loss=0.00569, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.22it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.72      0.78       302
    positive       0.94      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.84      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 32: 100%|██████████| 986/986 [00:12<00:00, 76.56it/s, loss=0.00597, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.69it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.72      0.78       302
    positive       0.94      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 33: 100%|██████████| 986/986 [00:12<00:00, 75.90it/s, loss=0.00625, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.20it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.72      0.78       302
    positive       0.94      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 34: 100%|██████████| 986/986 [00:12<00:00, 76.25it/s, loss=0.00642, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.32it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.82      0.76      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.86      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 35: 100%|██████████| 986/986 [00:12<00:00, 76.30it/s, loss=0.00417, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.26it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.88      0.65      0.75       302
    positive       0.93      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.91      0.82      0.85      1752
weighted avg       0.92      0.93      0.92      1752
```

```
训练中 36: 100%|██████████| 986/986 [00:12<00:00, 76.57it/s, loss=0.0047, acc
=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.12it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.83      0.75      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.86      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 37: 100%|██████████| 986/986 [00:12<00:00, 76.58it/s, loss=0.00403, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.69it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.84      0.75      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.86      0.88      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 38: 100%|██████████| 986/986 [00:12<00:00, 76.84it/s, loss=0.00436, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.06it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.84      0.75      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.86      0.88      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 39: 100%|██████████| 986/986 [00:12<00:00, 76.34it/s, loss=0.00365, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 100.29it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.83      0.77      0.80       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.87      0.88      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 40: 100%|██████████| 986/986 [00:12<00:00, 76.27it/s, loss=0.00343, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 100.11it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.84      0.73      0.78       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.89      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 41: 100%|██████████| 986/986 [00:12<00:00, 76.12it/s, loss=0.00361, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.92it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.86      0.71      0.78       302
    positive       0.94      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.84      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 42: 100%|██████████| 986/986 [00:12<00:00, 76.24it/s, loss=0.00429, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.56it/s]
```

```
Classification Report:
               precision    recall  f1-score   support

    negative       0.85      0.73      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 43: 100%|██████████| 986/986 [00:12<00:00, 76.37it/s, loss=0.00309, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.51it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.74      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.86      0.88      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 44: 100%|██████████| 986/986 [00:12<00:00, 76.43it/s, loss=0.00176, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 104.69it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.90      0.67      0.77       302
    positive       0.93      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.92      0.83      0.86      1752
weighted avg       0.93      0.93      0.93      1752
```

```
训练中 45: 100%|██████████| 986/986 [00:12<00:00, 76.42it/s, loss=0.00266, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 101.18it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.76      0.80       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.87      0.88      1752
weighted avg       0.93      0.93      0.93      1752
```

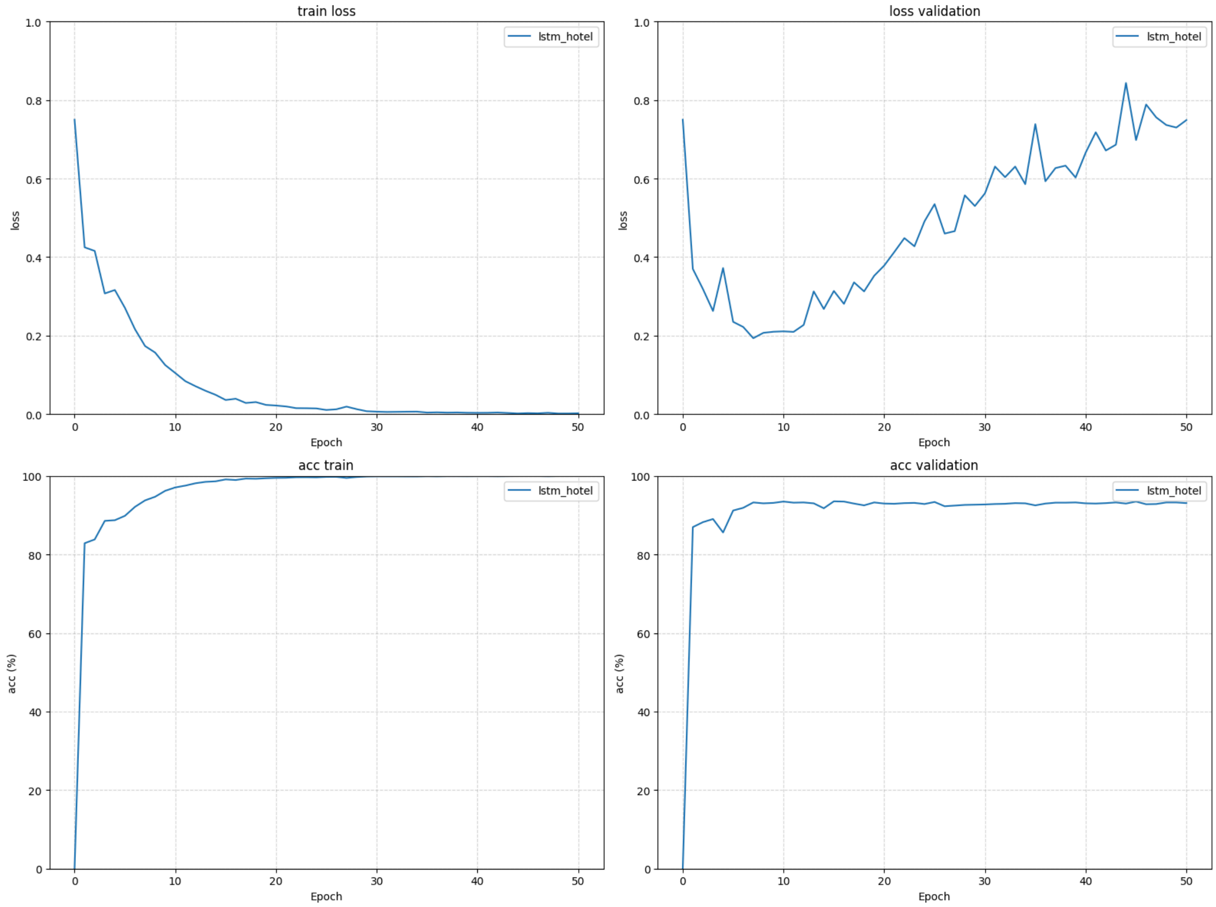
```
训练中 46: 100%|██████████| 986/986 [00:12<00:00, 76.49it/s, loss=0.00212, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 101.98it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.87      0.69      0.77       302
    positive       0.94      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.83      0.86      1752
weighted avg       0.93      0.93      0.92      1752
```

```
训练中 47: 100%|██████████| 986/986 [00:12<00:00, 76.38it/s, loss=0.00356, ac
c=99.9]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 103.35it/s]
```

```
Classification Report:
              precision    recall  f1-score   support

    negative       0.86      0.71      0.77       302
    positive       0.94      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.84      0.87      1752
weighted avg       0.93      0.93      0.93      1752

训练中 48: 100%|██████████| 986/986 [00:12<00:00, 76.22it/s, loss=0.00166, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 105.11it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.86      0.73      0.79       302
    positive       0.95      0.98      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752

训练中 49: 100%|██████████| 986/986 [00:12<00:00, 76.32it/s, loss=0.00167, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 102.86it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.74      0.79       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752

训练中 50: 100%|██████████| 986/986 [00:13<00:00, 71.79it/s, loss=0.00218, ac
c=100]
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 99.80it/s]
Classification Report:
              precision    recall  f1-score   support

    negative       0.85      0.73      0.78       302
    positive       0.95      0.97      0.96      1450

    accuracy                           0.93      1752
   macro avg       0.90      0.85      0.87      1752
weighted avg       0.93      0.93      0.93      1752


Saving model...distil_lstm_hotel.pth
```

In [81]:
```
plot_training_curves2(history_hotel_lstm, 'lstm_hotel')
```



In [71]:
```
model_lstm
```

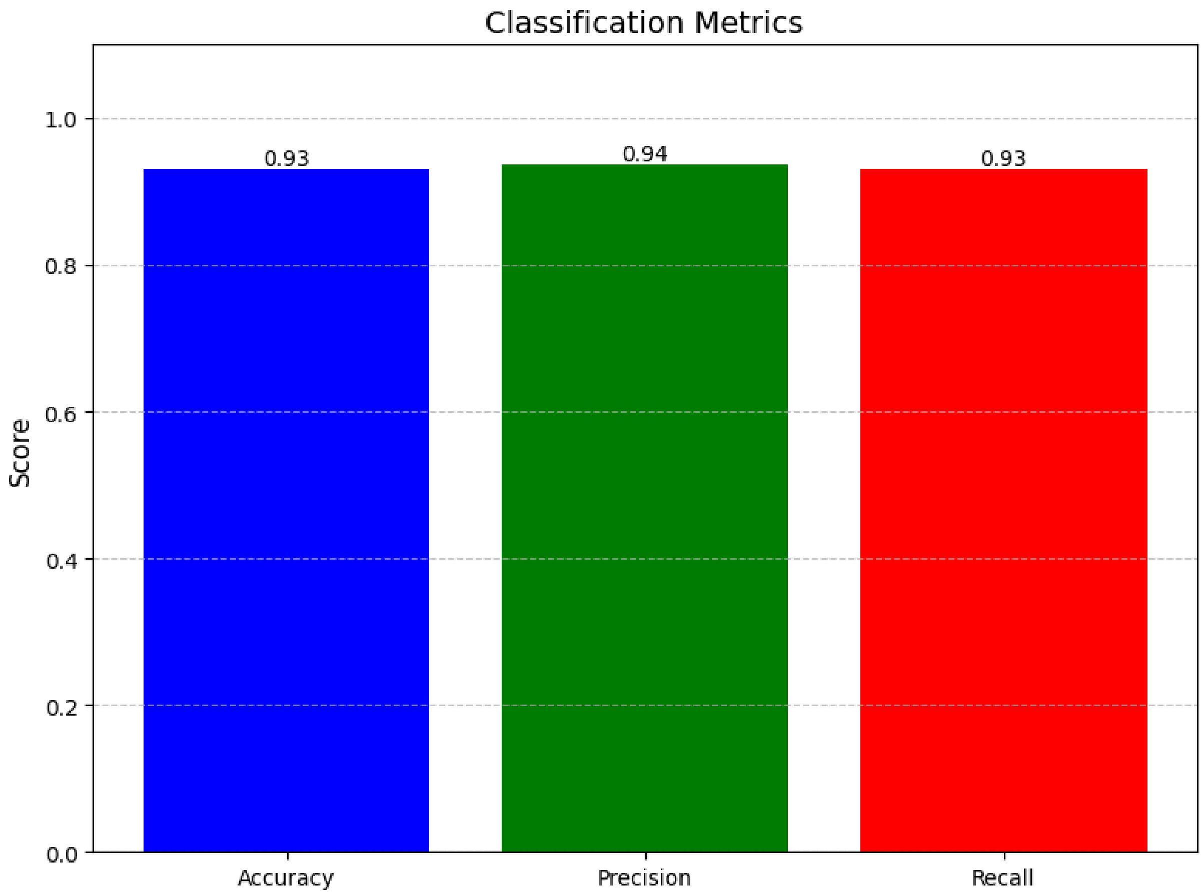
Out[71]:
```
BiLSTM_Attn(
  (embedding): Embedding(30522, 32, padding_idx=0)
  (lstm): LSTM(32, 32, batch_first=True, bidirectional=True)
  (fc): Linear(in_features=64, out_features=2, bias=True)
  (dropout): Dropout(p=0.7, inplace=False)
)
```

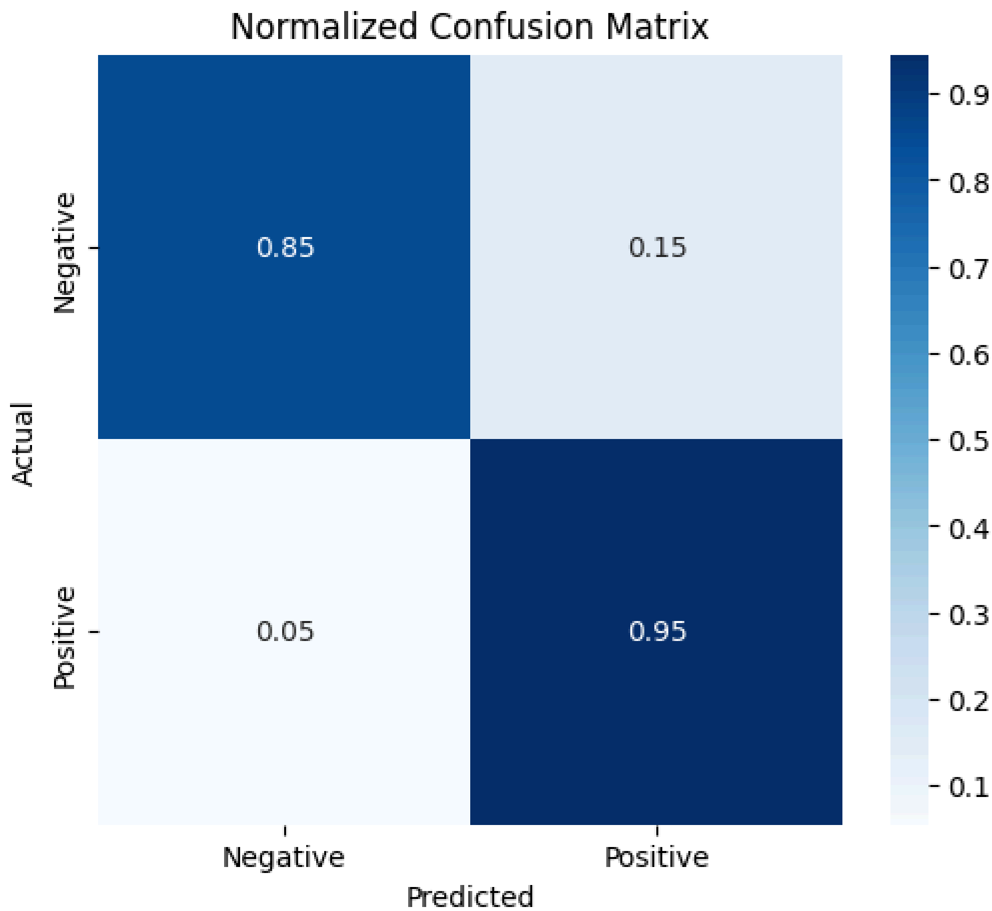
In [78]:
```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model_lstm.to(device)

test_true, test_preds, _, _ = evaluate2(model_lstm, device, test_loader_cnn)
accuracy = accuracy_score(test_true, test_preds)
print(f"Test Accuracy: {accuracy * 100:.2f}%\n")
```

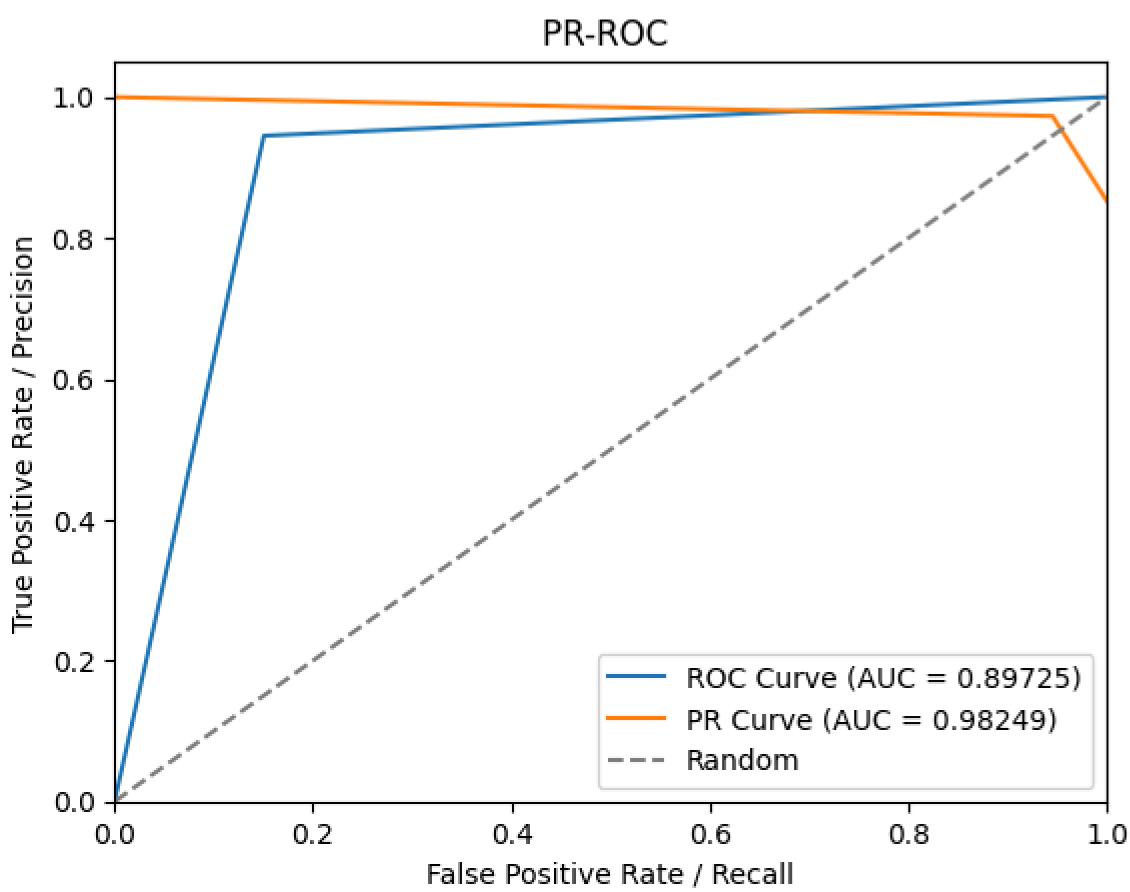
```
Evaluating: 100%|██████████| 110/110 [00:01<00:00, 106.31it/s]
Test Accuracy: 93.09%
```

In [79]:
```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f

acc = accuracy_score(test_true, test_preds)
precision = precision_score(test_true, test_preds, average='weighted')  # 或
recall = recall_score(test_true, test_preds, average='weighted')
f1 = f1_score(test_true, test_preds, average='weighted')
draw_bar(acc, precision, recall)
```

```
plot_confusion_matrix(test_true, test_preds, labels=["Negative", "Positive"]
plot_auc(test_preds, test_true)
```




Normalized Confusion Matrix
Actual
Negative
Positive
0.85
0.15
0.05
0.95
Predicted
Negative
Positive
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1


PR-ROC
True Positive Rate / Precision
False Positive Rate / Recall
ROC Curve (AUC = 0.89725)
PR Curve (AUC = 0.98249)
Random
0.0
0.2
0.4
0.6
0.8
1.0

Naive Bayes

In [2]:
```
param_grid = {
        # 'fit_prior': Categorical([True, False]),
    'alpha': [0.05, 0.1, 0.5, 1.0, 2.0, 3.0]

}
nb_model = MultinomialNB()
grid_search_model(nb_model, X_train_count, X_test_count,
                  y_train_count, y_test_count, param_grid=param_grid)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[2], line 6
      1 param_grid = {
      2         # 'fit_prior': Categorical([True, False]),
      3     'alpha': [0.05, 0.1, 0.5, 1.0, 2.0, 3.0]
      4
      5 }
----> 6 nb_model = MultinomialNB()
      7 grid_search_model(nb_model, X_train_count, X_test_count,
      8                   y_train_count, y_test_count, param_grid=param_grid)

NameError: name 'MultinomialNB' is not defined
```